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United States Patent Application Publication

20250279920

Kind Code

A1

Publication Date

September 04, 2025

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LONG TRAINING FIELD WITH REDUCED PEAK-TO-AVERAGE POWER RATIO

Abstract

Various aspects relate generally to the generation and wireless transmission of a long training field (LTF) and a data field within a 320 MHz or 240 MHz bandwidth channel. The LTF includes an LTF sequence based on a concatenation of a plurality of LTF subsequences, each LTF subsequence being associated with a channel bandwidth less than the total bandwidth of the channel, each of the plurality of LTF subsequences being selectively phase-rotated to reduce a peak-to-average power ratio. Each of the plurality of LTF subsequences may be associated with an LTF designed for an 80 MHz bandwidth channel. The LTF and the data field may then be transmitted in a wireless packet to at least one second wireless communication device via the 320 MHz or 240 MHz bandwidth channel.

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Family ID: 79022156

Appl. No.: 19/082962

Filed: March 18, 2025

Related U.S. Application Data

parent US continuation 17353673 20210621 parent-grant-document US 12284067 child US 19082962

us-provisional-application US 63042557 20200622

us-provisional-application US 63046569 20200630

us-provisional-application US 63051264 20200713

Publication Classification

Int. Cl.: H04L27/26 (20060101); H04W84/12 (20090101)

U.S. Cl.:

CPC H04L27/2621 (20130101); H04L27/2607 (20130101); H04L27/2613 (20130101);
H04W84/12 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This present Application for Patent is a Continuation of U.S. Non-Provisional patent application Ser. No. 17/353,673, by Wu et al., entitled “LONG TRAINING FIELD WITH REDUCED PEAK-TO-AVERAGE POWER RATIO,” filed Jun. 21, 2021, which also claims priority to and the benefit of U.S. provisional patent application No. 63/042,557 filed on Jun. 22, 2020, U.S. provisional patent application No. 63/046,569 filed on Jun. 30, 2020, and U.S. provisional patent application No. 63/051,264 filed on Jul. 13, 2020, the entire contents of which are incorporated herein by reference as if fully set forth below in their entirety and for all applicable purposes.

TECHNICAL FIELD

[0002] The present disclosure relates generally to wireless communication, and more particularly, to training fields for communications over large bandwidth channels that may achieve reduced peak-to-average power ratios.

DESCRIPTION OF THE RELATED TECHNOLOGY

[0003] As wireless communications have been evolving toward ever increasing data rates, the Institute of Electrical and Electronic Engineers (IEEE) has been evolving its IEEE 802.11 standard to provide increased throughput. Recently, IEEE 802.11be is being developed, which defines Extreme High Throughput (EHT) wireless communications using large bandwidth channels (for example, having a bandwidth of 240 MHz, 320 MHz, or larger). The total channel bandwidth may be comprised of a combination of subchannels (potentially having different sizes) in one or more frequency bands (such as the 5 GHz or 6 GHz frequency bands). The subchannels, which may be contiguous or non-contiguous in the frequency band, may be collectively referred to as a wireless channel. IEEE 802.11be proposes to transmit signals using orthogonal frequency-division multiple access (OFDMA) which is a multi-user version of the orthogonal frequency-division multiplexing (OFDM) digital modulation scheme. OFDM employs multi-carrier modulation where a plurality of carriers (for example, parallel sub-carriers), each carrying low bit rate data, are orthogonal to each other. In OFDMA, multiple access is achieved by assigning subsets of subcarriers to individual users or devices.

[0004] OFDMA transmissions can have high peak values in the time domain since many subcarrier components are added via an inverse fast Fourier transformation (IFFT) operation. Consequently, OFDMA transmissions may have a high peak-to-average power ratio (PAPR) when compared to single-carrier transmissions. The high PAPR is one of the most detrimental aspects in an OFDMA system as it decreases the signal-to-quantization noise ratio (SQNR) of the analog-digital convertor (ADC) and digital-analog convertor (DAC) while degrading the efficiency of the power amplifier in the transmitter. According to the IEEE 802.11 standards, a physical layer preamble of a wireless communication transmission includes one or more long training field (LTF). Each LTF includes a training sequence that can be used by a receiving device to set or adjust an automatic gain control (AGC) function of the receiving device. AGC is a technique in an amplifier or chain of amplifiers

used to adjust the signal amplitude at its output regardless of the variation of the signal amplitude at the input. The LTF may include a training sequence of predetermined signals that can be used to identify the beginning of the transmission and to set the AGC. However, a high PAPR tends to skew the AGC incorrectly, making it difficult to demodulate/detect control or data channels.

BRIEF SUMMARY OF SOME EXAMPLES

[0005] The following presents a simplified summary of one or more aspects of the present disclosure, in order to provide a basic understanding of such aspects. This summary is not an extensive overview of all contemplated features of the disclosure, and is intended neither to identify key or critical elements of all aspects of the disclosure nor to delineate the scope of any or all aspects of the disclosure. Its sole purpose is to present some concepts of one or more aspects of the disclosure in a simplified form as a prelude to the more detailed description that is presented later.

[0006] One aspect provides a method for wireless communication by a wireless communication device. A data field for transmission is generated using a first channel having a 320 MHz or 240 MHz total channel bandwidth. A long training field (LTF) sequence is obtained that is based on a concatenation of a plurality of LTF subsequences, each LTF subsequence being associated with a channel bandwidth less than the total channel bandwidth of the first channel, one or more of the plurality of LTF subsequences being selectively phase-rotated to reduce a peak-to-average power ratio. A wireless packet is then transmitted that includes the LTF and the data field to at least one second wireless communication device via the first channel. Each of the plurality of LTF subsequences may be based on a High Efficiency (HE) LTF (HE-LTF) sequence for a subchannel having an 80 MHz channel bandwidth.

[0007] In one example, a symbol duration of each symbol of the LTF sequence may be one-fourth as long as a symbol duration of each symbol of the data field. For instance, the symbol duration of each symbol of the LTF sequence may be $3.2\ \mu\text{s}$ plus a guard interval and the symbol duration of each symbol of the data field may be $12.8\ \mu\text{s}$ plus the guard interval. The concatenation may be a sequence formed by concatenating, in any order, a first LTF subsequence of the plurality of LTF subsequences, a second LTF subsequence of the plurality of LTF subsequences, a third LTF subsequence of the plurality of LTF subsequences, and a fourth LTF subsequence of the plurality of LTF subsequences, with k filler zeros between adjacent ones of the plurality of LTF subsequences. The selectively phase-rotated one or more of the plurality of LTF subsequences may be based on a set of phase rotation values $s=[s_1\ s_2\ s_3\ s_4\ s_5\ s_6\ s_7\ s_8]$ to the plurality of LTF subsequences, and each of the LTF subsequences is based on a first LTF base sequence LTF-A and a second LTF base sequence LTF-B. In one example, the first LTF subsequence= $\{s_1*\text{LTF-A}, 0, s_2*\text{LTF-B}\}$; the second LTF subsequence= $\{s_3*\text{LTF-A}, 0, s_4*\text{LTF-B}\}$; the third LTF subsequence= $\{s_5*\text{LTF-A}, 0, s_6*\text{LTF-B}\}$; and the fourth LTF subsequence= $\{s_7*\text{LTF-A}, 0, s_8*\text{LTF-B}\}$, where the set of phase rotation values $s=[s_1\ s_2\ s_3\ s_4\ s_5\ s_6\ s_7\ s_8]=[1\ 1\ 1\ 1\ -1\ -1\ -1\ -1]$.

[0008] In another example, a symbol duration of each symbol of the LTF sequence may be half as long as a symbol duration of each symbol of the data field. For instance, the symbol duration of each symbol of the LTF sequence may be $6.4\ \mu\text{s}$ plus a guard interval and the symbol duration of each symbol of the data field may be $12.8\ \mu\text{s}$ plus the guard interval. The concatenation may be a sequence formed by concatenating, in any order, a first LTF subsequence of the plurality of subsequences, a second LTF subsequence of the plurality of LTF subsequences, a third LTF subsequence of the plurality of LTF subsequences, and a fourth LTF subsequence of the plurality of LTF subsequences, with k filler zeros between adjacent ones of the plurality of LTF subsequences. The selectively phase-rotated one or more of the plurality of LTF subsequences may be based on a set of phase rotation values $q=[q_1\ q_2\ q_3\ q_4\ q_5\ q_6\ q_7\ q_8\ q_9\ q_{10}\ q_{11}\ q_{12}\ q_{13}\ q_{14}\ q_{15}\ q_{16}\ q_{17}\ q_{18}\ q_{19}\ q_{20}]$ to the plurality of LTF subsequences, and where each of the LTF subsequences is based on a first LTF base sequence LTF-A, a second LTF base sequence LTF-B, a third LTF base sequence LTF-C, a fourth LTF base sequence LTF-D, and a fifth LTF base sequence LTF-E. In one

example, the first LTF subsequence= $\{q1*\text{LTF-A}, q2*\text{LTF-B}, q3*\text{LTF-C}, q4*\text{LTF-D}, q5*\text{LTF-E}\}$, the second LTF subsequence= $\{q6*\text{LTF-A}, q7*\text{LTF-B}, q8*\text{LTF-C}, q9*\text{LTF-D}, q10*\text{LTF-E}\}$, the third LTF subsequence= $\{q11*\text{LTF-A}, q12*\text{LTF-B}, q13*\text{LTF-C}, q14*\text{LTF-D}, q15*\text{LTF-E}\}$, and the fourth LTF subsequence= $\{q16*\text{LTF-A}, q17*\text{LTF-B}, q18*\text{LTF-C}, q19*\text{LTF-D}, q20*\text{LTF-E}\}$, where the set of phase rotation values $q=[q1\ q2\ q3\ q4\ q5\ q6\ q7\ q8\ q9\ q10\ q11\ q12\ q13\ q14\ q15\ q16\ q17\ q18\ q19\ q20]=[+1\ -1\ -1\ -1\ +1\ -1\ +1\ +1\ +1\ -1\ -1\ +1\ +1\ +1\ -1\ -1\ +1\ -1\ -1\ +1]$ or $[+1\ +1\ -1\ -1\ -1\ -1\ -1\ -1\ +1\ +1\ +1\ +1\ +1\ +1\ +1\ +1\ +1]$.

[0009] In yet another example, a symbol duration of each symbol of the LTF sequence may be the same as a symbol duration of each symbol of the data field. For instance, the symbol duration of each symbol of the LTF sequence may be 12.8 μs plus a guard interval and the symbol duration of each symbol of the data field may be 12.8 μs plus the guard interval. The concatenation may be a sequence formed by concatenating, in any order, a first LTF subsequence of the plurality of LTF subsequences, a second LTF subsequence of the plurality of LTF subsequences, a third LTF subsequence of the plurality of LTF subsequences, and a fourth LTF subsequence of the plurality of LTF subsequences, with k filler zeros between adjacent ones of the plurality of LTF subsequences. The selectively phase-rotated one or more of the plurality of LTF subsequences may be based on a set of phase rotation values $n=[n1\ n2\ n3\ n4\ n5\ n6\ n7\ n8\ n9\ n10\ n11\ n12\ n13\ n14\ n15\ n16]$ to the plurality of LTF subsequences, and each of the LTF subsequences being based on a first LTF base sequence LTF-A, a second LTF base sequence LTF-B, a third LTF base sequence LTF-C, and a fourth LTF base sequence LTF-D. In one example, the first LTF= $\{n1*\text{LTF-A}, n2*\text{LTF-B}, 0, n3*\text{LTF-C}, n4*\text{LTF-D}\}$; the second LTF= $\{n5*\text{LTF-A}, n6*\text{LTF-B}, 0, n7*\text{LTF-C}, n8*\text{LTF-D}\}$; the third LTF= $\{n9*\text{LTF-A}, n10*\text{LTF-B}, 0, n11*\text{LTF-C}, n12*\text{LTF-D}\}$; and the fourth LTF= $\{n13*\text{LTF-A}, n14*\text{LTF-B}, 0, n15*\text{LTF-C}, n16*\text{LTF-D}\}$, where the set of phase rotation values $n=[n1\ n2\ n3\ n4\ n5\ n6\ n7\ n8\ n9\ n10\ n11\ n12\ n13\ n14\ n15\ n16]=[1\ -1\ 1\ -1\ 1\ -1\ -1\ 1\ 1\ 1\ -1\ -1\ 1\ 1\ 1\ 1]$ or $[1\ -1\ -1\ 1\ 1\ -1\ 1\ -1\ 1\ 1\ 1\ 1\ 1\ -1\ -1]$.

[0010] In an alternative example, a symbol duration of each symbol of the LTF sequence may be the same as a symbol duration of each symbol of the data field. For instance, the symbol duration of each symbol of the LTF sequence may be 12.8 μs plus a guard interval and the symbol duration of each symbol of the data field may be 12.8 μs plus the guard interval. The concatenation may be a sequence formed by concatenating, in any order, a first LTF subsequence of the plurality of LTF subsequences, a second LTF subsequence of the plurality of LTF subsequences, a third LTF subsequence of the plurality of LTF subsequences, and a fourth LTF subsequence of the plurality of LTF subsequences, with k filler zeros between adjacent ones of the plurality of LTF subsequences. The selectively phase-rotated one or more of the plurality of LTF subsequences may be based on a set of phase rotation values $n=[n1\ n2\ n3\ n4\ n5\ n6\ n7\ n8]$ to the plurality of LTF subsequences, each of the LTF subsequences being based on a first LTF base sequence LTF-A and a second LTF base sequence LTF-B. In one example, the first LTF subsequence= $\{n1*\text{LTF-A}, 0, n2*\text{LTF-B}\}$; the second LTF subsequence= $\{n3*\text{LTF-A}, 0, n4*\text{LTF-B}\}$; the third LTF subsequence= $\{n5*\text{LTF-A}, 0, n6*\text{LTF-B}\}$; and the fourth LTF subsequence= $\{n7*\text{LTF-A}, 0, n8*\text{LTF-B}\}$, where the set of phase rotation values $n=[n1\ n2\ n3\ n4\ n5\ n6\ n7\ n8]=[1\ -1\ -1\ -1\ -1\ 1\ 1\ 1]$.

[0011] In yet another example, each of the plurality of LTF subsequences may be based on an 80 MHz LTF segment formed from two High Efficiency (HE) LTF (HE-LTF) sequences for subchannels having a 40 MHz channel bandwidth. The 80 MHz LTF segment may be formed by interleaving and phase rotating values of four base sub-segments $S_a, S_b, S_c,$ and S_d , where [0012] $S_a=[+1\ +1\ +1\ +1\ +1\ -1\ -1\ +1\ +1\ -1\ +1\ -1\ +1]$, [0013] $S_b=[S_a(1:7), -S_a(8:13)]$, [0014] $S_c=[-S_a(1:6), S_a(7:13)]$, and [0015] $S_d=[S_d(1:14)]$, where the values $S_d(x)$ are configured to reduce a peak-to-average power ratio in an 80 MHz segment. In one example, $S_d=[-1, -1, -1, -1, 1, -1, -1, -1, 1, 1, 1, -1, -1, -1]$.

[0016] The symbol duration of each symbol of the LTF sequence may be half as long as a symbol duration of each symbol of the data field, and the concatenation may be a sequence formed by

concatenating, in any order, a first LTF subsequence of the plurality of subsequences, a second LTF subsequence of the plurality of LTF subsequences, a third LTF subsequence of the plurality of LTF subsequences, and a fourth LTF subsequence of the plurality of LTF subsequences, with k filler zeros between adjacent ones of the plurality of LTF subsequences. The symbol duration of each symbol of the LTF sequence may be $6.4\ \mu\text{s}$ plus a guard interval and the symbol duration of each symbol of the data field may be $12.8\ \mu\text{s}$ plus the guard interval. The selectively phase-rotated one or more of the plurality of LTF subsequences may be based on a set of phase rotation values $q=[q_1\ q_2\ q_3\ q_4\ q_5\ q_6\ q_7\ q_8]$ applied to the plurality of LTF subsequences, each of the LTF subsequences being based on a first LTF base sequence LTF-A and a second LTF base sequence LTF-B. In one example, the first LTF subsequence= $\{q_1*\text{LTF-A}, 0, q_2*\text{LTF-B}\}$, the second LTF subsequence= $\{q_3*\text{LTF-A}, 0, q_4*\text{LTF-B}\}$, the third LTF subsequence= $\{q_5*\text{LTF-A}, 0, q_6*\text{LTF-B}\}$, and the fourth LTF subsequence= $\{q_7*\text{LTF-A}, 0, q_8*\text{LTF-B}\}$, where the set of phase rotation values $q=[q_1\ q_2\ q_3\ q_4\ q_5\ q_6\ q_7\ q_8]=[1\ 1\ 1\ 1\ -1\ -1\ -1\ 1]$.

[0017] Yet another aspect provides a method for receiving wireless communications. In particular, a wireless packet that includes a long training field (LTF) sequence and a data field may be received via a first channel having a 320 MHz or 240 MHz total channel bandwidth. The LTF sequence may then be detected within the packet, the LTF sequence based on a concatenation of a plurality of LTF subsequences, each LTF subsequence being associated with a channel bandwidth less than the total channel bandwidth of the first channel, one or more of the plurality of LTF subsequences being selectively phase-rotated to reduce a peak-to-average power ratio. The data field may be received or obtained based on the detected LTF sequence.

[0018] These and other aspects of the present disclosure will become more fully understood upon a review of the detailed description, which follows. Other aspects, features, and examples will become apparent to those of ordinary skill in the art, upon reviewing the following description of specific, example implementations in conjunction with the accompanying figures. While features of the present examples may be discussed relative to certain implementations and figures below, all implementations can include one or more of the advantageous features discussed herein. In other words, while one or more implementations may be discussed as having certain advantageous features, one or more of such features may also be used in accordance with the various implementations discussed herein. In similar fashion, while example implementations may be discussed below as a device, system, or method it should be understood that such example implementations can be implemented in various devices, systems, and methods.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] FIG. 1 shows a block diagram of an example wireless communication network.

[0020] FIG. 2A shows an example protocol data unit (PDU) usable for wireless communication between an access point and one or more stations.

[0021] FIG. 2B shows an example L-SIG in the PDU of FIG. 2A.

[0022] FIG. 3A shows an example physical layer convergence protocol PDU (PPDU) usable for wireless communication between an access point and one or more stations.

[0023] FIG. 3B shows another example PPDU usable for wireless communication between an access point and one or more stations.

[0024] FIG. 4 shows an example 2N-tone plan.

[0025] FIG. 5 shows example modes available for 320 MHz bandwidth transmissions and 240 MHz bandwidth transmissions.

[0026] FIG. 6 illustrates examples of OFDMA resource unit (RU) allocations.

[0027] FIG. 7 illustrates examples of tone spacings and index ranges for various fast Fourier

transform (FFT) sizes and symbol durations for 80 MHz, 160 MHz, and 320 MHz transmissions.

[0028] FIG. **8** illustrates an example of an Extreme High Throughput (EHT) long training field (LTF) (EHT-LTF) construction for a 320 MHz bandwidth channel, where each symbol of the EHT-LTF uses a $1\times$ symbol duration and each symbol of a data field uses a $4\times$ symbol duration.

[0029] FIG. **9** illustrates examples of High Efficiency (HE) LTF (HE-LTF) subsequences for an 80 MHz bandwidth channel including a first LTF base sequence and a second LTF base sequence.

[0030] FIG. **10** illustrates an example of an EHT-LTF construction for a 320 MHz bandwidth channel, where each symbol of the EHT-LTF uses a $2\times$ symbol duration and each symbol of a data field uses a $4\times$ symbol duration.

[0031] FIG. **11** illustrates examples of HE-LTF subsequences for an 80 MHz bandwidth channel including a first LTF base sequence, a second LTF base sequence, a third LTF base sequence, a fourth LTF base sequence, and a fifth LTF base sequence.

[0032] FIG. **12** illustrates another example of an EHT-LTF construction for a 320 MHz bandwidth channel, where each symbol of the EHT-LTF uses a $2\times$ symbol duration and each symbol of a data field uses a $4\times$ symbol duration.

[0033] FIG. **13** illustrates examples of HE-LTF subsequences for an 80 MHz bandwidth channel including a first LTF base sequence LTF-A and a second LTF base sequence LTF-B.

[0034] FIG. **14** illustrates a first example of an EHT long training field (LTF) construction for a 320 MHz bandwidth channel, where each symbol of the EHT-LTF uses a $4\times$ symbol duration and each symbol of a data field uses a $4\times$ symbol duration.

[0035] FIG. **15** illustrates examples of HE-LTF subsequences for an 80 MHz bandwidth channel including a first LTF base sequence, a second LTF base sequence, a third base sequence, and a fourth base sequence.

[0036] FIG. **16** illustrates a second example of an EHT long training field (LTF) construction for a 320 MHz bandwidth channel, where each symbol of the EHT-LTF uses a $4\times$ symbol duration and each symbol of a data field uses a $4\times$ symbol duration.

[0037] FIG. **17** illustrates examples of HE-LTF subsequences for an 80 MHz bandwidth channel including a first LTF base sequence and a second LTF base sequence.

[0038] FIG. **18** is a block diagram illustrating an example of a wireless communication device that supports training fields for communications over large bandwidth channels that may achieve reduced peak-to-average power ratios.

[0039] FIG. **19** is a flow chart illustrating an example method operational at a wireless communication device that supports training fields for communications over large bandwidth channels that may achieve reduced peak-to-average power ratios.

[0040] FIG. **20** is a flow chart illustrating an example method operational at a wireless communication device that supports training fields for communications over large bandwidth channels that may achieve reduced peak-to-average power ratios.

DETAILED DESCRIPTION

[0041] The following description is directed to some particular examples for the purposes of describing innovative aspects of this disclosure. However, a person having ordinary skill in the art will readily recognize that the teachings herein can be applied in a multitude of different ways. Some or all of the described examples may be implemented in any device, system or network that is capable of transmitting and receiving radio frequency (RF) signals according to one or more of the Institute of Electrical and Electronics Engineers (IEEE) 802.11 standards, the IEEE 802.15 standards, the Bluetooth® standards as defined by the Bluetooth Special Interest Group (SIG), or the Long Term Evolution (LTE), 3G, 4G or 5G (New Radio (NR)) standards promulgated by the 3rd Generation Partnership Project (3GPP), among others. The described implementations can be implemented in any device, system or network that is capable of transmitting and receiving RF signals according to one or more of the following technologies or techniques: code division multiple access (CDMA), time division multiple access (TDMA), frequency division multiple

access (FDMA), orthogonal FDMA (OFDMA), single-carrier FDMA (SC-FDMA), single-user (SU) multiple-input multiple-output (MIMO) and multi-user (MU)-MIMO. The described implementations also can be implemented using other wireless communication protocols or RF signals suitable for use in one or more of a wireless personal area network (WPAN), a wireless local area network (WLAN), a wireless wide area network (WWAN), or an internet of things (IOT) network.

[0042] As the IEEE 802.11 standards evolve to use higher bandwidths (for example, 320 MHz for Extreme High Throughput (EHT) operation as defined by the IEEE 802.11be amendment), it has been recognized that OFDMA transmissions can have high peak values in the time domain since many subcarrier components are added via an inverse fast Fourier transformation (IFFT) operation. Consequently, OFDMA transmissions may have a high peak-to-average power ratio (PAPR) when compared to single-carrier transmissions. The high PAPR of OFDMA transmissions decreases the signal-to-quantization noise ratio (SQNR) of the analog-digital convertor (ADC) and digital-analog convertor (DAC) in a radio chain while degrading the efficiency of the power amplifier in a transmitter.

[0043] According to the IEEE 802.11 standards, a physical layer preamble of a wireless communication transmission includes one or more long training fields (LTFs). For example, an EHT transmission includes a legacy LTF (L-LTF) and an EHT LTF (EHT-LTF). Each of the LTFs includes a training sequence of predetermined signals that can be used by a receiving device to identify the beginning of a transmission and for various other purposes. For instance, the EHT LTF may be specifically configured for channel estimation, automatic gain control (AGC), and other operations. This enables a receiving device to perform coherent demodulation/detection of the control or data channels.

[0044] Various aspects relate generally to an LTF constructed to minimize or reduce the peak-to-average-power ratio (PAPR) of the LTF. Some aspects more specifically relate to reutilizing a High Efficiency (HE)-LTF sequence defined by the IEEE 802.11ax amendment for 80 MHz bandwidth channels as a basis to construct a longer EHT-LTF sequence for 320 MHz bandwidth channels. In order to minimize the PAPR of the EHT-LTF for 320 MHz bandwidth channels, the HE-LTF for 80 MHz bandwidth channels may be modified based on phase rotations and concatenations to achieve a minimized PAPR or otherwise a PAPR below a threshold level. One or more combinations of phase rotation coefficients and HE-LTF segments that provide such minimized or reduced PAPR may be predetermined or preconfigured. In another aspect, an EHT-LTF sequence may be optimized to reduce the PAPR based on multi-stream PAPR minimization for different resource unit (RU) or multi-RU (MRU) sizes.

[0045] Particular aspects of the subject matter described in this disclosure can be implemented to realize one or more of the following potential advantages. For large bandwidth channels using OFDMA modulation, it is advantageous to configure the LTF to minimize or reduce the peak-to-average-power ratio (PAPR) of the LTF so that the LTF does not skew the AGC incorrectly. This allows a receiving device to perform coherent demodulation/detection of control or data channels. In one aspect, such LTF may be pre-generated and stored, so that it does not have to be dynamically generated every time.

[0046] FIG. 1 shows a block diagram of an example wireless communication network **100**.

According to some aspects, the wireless communication network **100** can be an example of a wireless local area network (WLAN) such as a Wi-Fi network (and will hereinafter be referred to as WLAN **100**). For example, the WLAN **100** can be a network implementing at least one of the IEEE 802.11 family of wireless communication protocol standards (such as that defined by the IEEE 802.11-2016 specification or amendments thereof including, but not limited to, 802.11ay, 802.11ax, 802.11az, 802.11ba and 802.11be). The WLAN **100** may include numerous wireless communication devices such as an access point (AP) **102** and multiple stations (STAs) **104**. While only one AP **102** is shown, the WLAN network **100** also can include multiple APs **102**.

[0047] Each of the STAs **104** also may be referred to as a mobile station (MS), a mobile device, a mobile handset, a wireless handset, an access terminal (AT), a user equipment (UE), a subscriber station (SS), or a subscriber unit, among other examples. The STAs **104** may represent various devices such as mobile phones, personal digital assistant (PDAs), other handheld devices, netbooks, notebook computers, tablet computers, laptops, display devices (for example, TVs, computer monitors, navigation systems, among others), music or other audio or stereo devices, remote control devices (“remotes”), printers, kitchen or other household appliances, key fobs (for example, for passive keyless entry and start (PKES) systems), among other examples.

[0048] A single AP **102** and an associated set of STAs **104** may be referred to as a basic service set (BSS), which is managed by the respective AP **102**. FIG. **1** additionally shows an example coverage area **106** of the AP **102**, which may represent a basic service area (BSA) of the WLAN **100**. The BSS may be identified to users by a service set identifier (SSID), as well as to other devices by a basic service set identifier (BSSID), which may be a medium access control (MAC) address of the AP **102**. The AP **102** periodically broadcasts beacon frames (“beacons”) including the BSSID to enable any STAs **104** within wireless range of the AP **102** to “associate” or re-associate with the AP **102** to establish a respective communication link **108** (hereinafter also referred to as a “Wi-Fi link”), or to maintain a communication link **108**, with the AP **102**. For example, the beacons can include an identification of a primary channel used by the respective AP **102** as well as a timing synchronization function for establishing or maintaining timing synchronization with the AP **102**. The AP **102** may provide access to external networks to various STAs **104** in the WLAN via respective communication links **108**.

[0049] To establish a communication link **108** with an AP **102**, each of the STAs **104** is configured to perform passive or active scanning operations (“scans”) on frequency channels in one or more frequency bands (for example, the 2.4 GHz, 5 GHz, 6 GHz or 60 GHz bands). To perform passive scanning, a STA **104** listens for beacons, which are transmitted by respective APs **102** at a periodic time interval referred to as the target beacon transmission time (TBTT) (measured in time units (TUs) where one TU may be equal to **1024** microseconds (μ s)). To perform active scanning, a STA **104** generates and sequentially transmits probe requests on each channel to be scanned and listens for probe responses from APs **102**. Each STA **104** may be configured to identify or select an AP **102** with which to associate based on the scanning information obtained through the passive or active scans, and to perform authentication and association operations to establish a communication link **108** with the selected AP **102**. The AP **102** assigns an association identifier (AID) to the STA **104** at the culmination of the association operations, which the AP **102** uses to track the STA **104**.

[0050] As a result of the increasing ubiquity of wireless networks, a STA **104** may have the opportunity to select one of many BSSs within range of the STA or to select among multiple APs **102** that together form an extended service set (ESS) including multiple connected BSSs. An extended network station associated with the WLAN **100** may be connected to a wired or wireless distribution system that may allow multiple APs **102** to be connected in such an ESS. As such, a STA **104** can be covered by more than one AP **102** and can associate with different APs **102** at different times for different transmissions. Additionally, after association with an AP **102**, a STA **104** also may be configured to periodically scan its surroundings to find a more suitable AP **102** with which to associate. For example, a STA **104** that is moving relative to its associated AP **102** may perform a “roaming” scan to find another AP **102** having more desirable network characteristics such as a greater received signal strength indicator (RSSI) or a reduced traffic load.

[0051] In some cases, STAs **104** may form networks without APs **102** or other equipment other than the STAs **104** themselves. One example of such a network is an ad hoc network (or wireless ad hoc network). Ad hoc networks may alternatively be referred to as mesh networks or peer-to-peer (P2P) networks. In some cases, ad hoc networks may be implemented within a larger wireless network such as the WLAN **100**. In such implementations, while the STAs **104** may be capable of communicating with each other through the AP **102** using communication links **108**, STAs **104** also

can communicate directly with each other via direct wireless links **110**. Additionally, two STAs **104** may communicate via a direct communication link **110** regardless of whether both STAs **104** are associated with and served by the same AP **102**. In such an ad hoc system, one or more of the STAs **104** may assume the role filled by the AP **102** in a BSS. Such a STA **104** may be referred to as a group owner (GO) and may coordinate transmissions within the ad hoc network. Examples of direct wireless links **110** include Wi-Fi Direct connections, connections established by using a Wi-Fi Tunneled Direct Link Setup (TDLS) link, and other P2P group connections.

[0052] The APs **102** and STAs **104** may function and communicate (via the respective communication links **108**) according to the IEEE **802.11** family of wireless communication protocol standards (such as that defined by the IEEE 802.11-2016 specification or amendments thereof including, but not limited to, 802.11ay, 802.11ax, 802.11az, 802.11ba and 802.11be). These standards define the WLAN radio and baseband protocols for the PHY and medium access control (MAC) layers. The APs **102** and STAs **104** transmit and receive wireless communications (hereinafter also referred to as “Wi-Fi communications”) to and from one another in the form of PHY protocol data units (PPDUs) (or physical layer convergence protocol (PLCP) PDUs). The APs **102** and STAs **104** in the WLAN **100** may transmit PPDUs over an unlicensed spectrum, which may be a portion of spectrum that includes frequency bands traditionally used by Wi-Fi technology, such as the 2.4 GHz band, the 5 GHz band, the 60 GHz band, the 3.6 GHz band, and the 900 MHz band. Some implementations of the APs **102** and STAs **104** described herein also may communicate in other frequency bands, such as the 6 GHz band, which may support both licensed and unlicensed communications. The APs **102** and STAs **104** also can be configured to communicate over other frequency bands such as shared licensed frequency bands, where multiple operators may have a license to operate in the same or overlapping frequency band or bands.

[0053] Each of the frequency bands may include multiple sub-bands or frequency channels. For example, PPDUs conforming to the IEEE 802.11n, 802.11ac, 802.11ax and 802.11be standard amendments may be transmitted over the 2.4, 5 GHz or 6 GHz bands, each of which is divided into multiple 20 MHz channels. As such, these PPDUs are transmitted over a physical channel having a minimum bandwidth of 20 MHz, but larger channels can be formed through channel bonding. For example, PPDUs may be transmitted over physical channels having bandwidths of 40 MHz, 80 MHz, 160 or 320 MHz by bonding together multiple 20 MHz channels.

[0054] Each PPDU is a composite structure that includes a PHY preamble and a payload in the form of a PHY service data unit (PSDU). The information provided in the preamble may be used by a receiving device to decode the subsequent data in the PSDU. In instances in which PPDUs are transmitted over a bonded channel, the preamble fields may be duplicated and transmitted in each of the multiple component channels. The PHY preamble may include both a legacy portion (or “legacy preamble”) and a non-legacy portion (or “non-legacy preamble”). The legacy preamble may be used for packet detection, automatic gain control and channel estimation, among other uses. The legacy preamble also may generally be used to maintain compatibility with legacy devices. The format of, coding of, and information provided in the non-legacy portion of the preamble is based on the particular IEEE 802.11 protocol to be used to transmit the payload.

[0055] FIG. 2A shows an example protocol data unit (PDU) **200** usable for wireless communication between an AP **102** and one or more STAs **104**. For example, the PDU **200** can be configured as a PPDU. As shown, the PDU **200** includes a PHY preamble **202** and a PHY payload **204**. For example, the preamble **202** may include a legacy portion that itself includes a legacy short training field (L-STF) **206**, which may consist of two BPSK symbols, a legacy long training field (L-LTF) **208**, which may consist of two BPSK symbols, and a legacy signal field (L-SIG) **210**, which may consist of two BPSK symbols. The legacy portion of the preamble **202** may be configured according to the IEEE 802.11a wireless communication protocol standard. The preamble **202** may also include a non-legacy portion including one or more non-legacy fields **212**, for example, conforming to an IEEE wireless communication protocol such as the IEEE 802.11ac,

802.11ax, 802.11be or later wireless communication protocol protocols.

[0056] The L-STF **206** generally enables a receiving device to perform coarse timing and frequency tracking and automatic gain control (AGC). The L-LTF **208** generally enables a receiving device to perform fine timing and frequency tracking and also to perform an initial estimate of the wireless channel. The L-SIG **210** generally enables a receiving device to determine a duration of the PDU and to use the determined duration to avoid transmitting on top of the PDU. For example, the L-STF **206**, the L-LTF **208** and the L-SIG **210** may be modulated according to a binary phase shift keying (BPSK) modulation scheme. The payload **204** may be modulated according to a BPSK modulation scheme, a quadrature BPSK (Q-BPSK) modulation scheme, a quadrature amplitude modulation (QAM) modulation scheme, or another appropriate modulation scheme. The payload **204** may include a PSDU including a data field (DATA) **214** that, in turn, may carry higher layer data, for example, in the form of medium access control (MAC) protocol data units (MPDUs) or an aggregated MPDU (A-MPDU).

[0057] FIG. 2B shows an example L-SIG **210** in the PDU **200** of FIG. 2A. The L-SIG **210** includes a data rate field **222**, a reserved bit **224**, a length field **226**, a parity bit **228**, and a tail field **230**. The data rate field **222** indicates a data rate (note that the data rate indicated in the data rate field **212** may not be the actual data rate of the data carried in the payload **204**). The length field **226** indicates a length of the packet in units of, for example, symbols or bytes. The parity bit **228** may be used to detect bit errors. The tail field **230** includes tail bits that may be used by the receiving device to terminate operation of a decoder (for example, a Viterbi decoder). The receiving device may utilize the data rate and the length indicated in the data rate field **222** and the length field **226** to determine a duration of the packet in units of, for example, microseconds (μ s) or other time units.

[0058] FIG. 3A shows an example PPDU **300** usable for wireless communication between an AP and one or more STAs. The PPDU **300** may be used for SU, OFDMA or MU-MIMO transmissions. The PPDU **300** may be formatted as a High Efficiency (HE) WLAN PPDU in accordance with the IEEE 802.11ax amendment to the IEEE 802.11 wireless communication protocol standard. The PPDU **300** includes a PHY preamble including a legacy portion **302** and a non-legacy portion **304**. The PPDU **300** may further include a PHY payload **306** after the preamble, for example, in the form of a PSDU including a data field **324**.

[0059] The legacy portion **302** of the preamble includes an L-STF **308**, an L-LTF **310**, and an L-SIG **312**. The non-legacy portion **304** includes a repetition of L-SIG (RL-SIG) **314**, a first HE signal field (HE-SIG-A) **316**, an HE short training field (HE-STF) **320**, and one or more HE long training fields (or symbols) (HE-LTFs) **322**. For OFDMA or MU-MIMO communications, the second portion **304** further includes a second HE signal field (HE-SIG-B) **318** encoded separately from HE-SIG-A **316**. HE-STF **320** may be used for timing and frequency tracking and AGC, and HE-LTF **322** may be used for more refined channel estimation. Like the L-STF **308**, L-LTF **310**, and L-SIG **312**, the information in RL-SIG **314** and HE-SIG-A **316** may be duplicated and transmitted in each of the component 20 MHz channels in instances involving the use of a bonded channel. In contrast, the content in HE-SIG-B **318** may be unique to each 20 MHz channel and target specific STAs **104**.

[0060] RL-SIG **314** may indicate to HE-compatible STAs **104** that the PPDU **300** is an HE PPDU. An AP **102** may use HE-SIG-A **316** to identify and inform multiple STAs **104** that the AP has scheduled UL or DL resources for them. For example, HE-SIG-A **316** may include a resource allocation subfield that indicates resource allocations for the identified STAs **104**. HE-SIG-A **316** may be decoded by each HE-compatible STA **104** served by the AP **102**. For MU transmissions, HE-SIG-A **316** further includes information usable by each identified STA **104** to decode an associated HE-SIG-B **318**. For example, HE-SIG-A **316** may indicate the frame format, including locations and lengths of HE-SIG-Bs **318**, available channel bandwidths and modulation and coding schemes (MCSs), among other examples. HE-SIG-A **316** also may include HE WLAN signaling

information usable by STAs **104** other than the identified STAs **104**.

[0061] HE-SIG-B **318** may carry STA-specific scheduling information such as, for example, STA-specific (or “user-specific”) MCS values and STA-specific RU allocation information. In the context of DL MU-OFDMA, such information enables the respective STAs **104** to identify and decode corresponding resource units (RUs) in the associated data field **324**. Each HE-SIG-B **318** includes a common field and at least one STA-specific field. The common field can indicate RU allocations to multiple STAs **104** including RU assignments in the frequency domain, indicate which RUs are allocated for MU-MIMO transmissions and which RUs correspond to MU-OFDMA transmissions, and the number of users in allocations, among other examples. The common field may be encoded with common bits, CRC bits, and tail bits. The user-specific fields are assigned to particular STAs **104** and may be used to schedule specific RUs and to indicate the scheduling to other WLAN devices. Each user-specific field may include multiple user block fields. Each user block field may include two user fields that contain information for two respective STAs to decode their respective RU payloads in data field **324**.

[0062] FIG. **3B** shows another example PPDU **350** usable for wireless communication between an AP and one or more STAs. The PPDU **350** may be used for SU, OFDMA or MU-MIMO transmissions. The PPDU **350** may be formatted as an Extreme High Throughput (EHT) WLAN PPDU in accordance with the IEEE 802.11be amendment to the IEEE 802.11 wireless communication protocol standard, or may be formatted as a PPDU conforming to any later (post-EHT) version of a new wireless communication protocol conforming to a future IEEE 802.11 wireless communication protocol standard or other wireless communication standard. The PPDU **350** includes a PHY preamble including a legacy portion **352** and a non-legacy portion **354**. The PPDU **350** may further include a PHY payload **356** after the preamble, for example, in the form of a PSDU including a data field **374**.

[0063] The legacy portion **352** of the preamble includes an L-STF **358**, an L-LTF **360**, and an L-SIG **362**. The non-legacy portion **354** of the preamble includes an RL-SIG **364** and multiple wireless communication protocol version-dependent signal fields after RL-SIG **364**. For example, the non-legacy portion **354** may include a universal signal field **366** (referred to herein as “U-SIG **366**”) and an EHT signal field **368** (referred to herein as “EHT-SIG **368**”). One or both of U-SIG **366** and EHT-SIG **368** may be structured as, and carry version-dependent information for, other wireless communication protocol versions beyond EHT. The non-legacy portion **354** further includes an additional short training field **370** (referred to herein as “EHT-STF **370**,” although it may be structured as, and carry version-dependent information for, other wireless communication protocol versions beyond EHT) and one or more additional long training fields **372** (referred to herein as “EHT-LTFs **372**,” although they may be structured as, and carry version-dependent information for, other wireless communication protocol versions beyond EHT). EHT-STF **370** may be used for timing and frequency tracking and AGC, and EHT-LTF **372** may be used for more refined channel estimation. Like L-STF **358**, L-LTF **360**, and L-SIG **362**, the information in U-SIG **366** and EHT-SIG **368** may be duplicated and transmitted in each of the component 20 MHz channels in instances involving the use of a bonded channel. In some implementations, EHT-SIG **368** may additionally or alternatively carry information in one or more non-primary 20 MHz channels that is different than the information carried in the primary 20 MHz channel.

[0064] EHT-SIG **368** may include one or more jointly encoded symbols and may be encoded in a different block from the block in which U-SIG **366** is encoded. EHT-SIG **368** may be used by an AP to identify and inform multiple STAs **104** that the AP has scheduled UL or DL resources for them. EHT-SIG **368** may be decoded by each compatible STA **104** served by the AP **102**. EHT-SIG **368** may generally be used by a receiving device to interpret bits in the data field **374**. For example, EHT-SIG **368** may include RU allocation information, spatial stream configuration information, and per-user signaling information such as MCSs, among other examples. EHT-SIG **368** may further include a cyclic redundancy check (CRC) (for example, four bits) and a tail (for example, 6

bits) that may be used for binary convolutional code (BCC). In some implementations, EHT-SIG **368** may include one or more code blocks that each include a CRC and a tail. In some aspects, each of the code blocks may be encoded separately.

[0065] EHT-SIG **368** may carry STA-specific scheduling information such as, for example, user-specific MCS values and user-specific RU allocation information. EHT-SIG **368** may generally be used by a receiving device to interpret bits in the data field **374**. In the context of DL MU-OFDMA, such information enables the respective STAs **104** to identify and decode corresponding RUs in the associated data field **374**. Each EHT-SIG **368** may include a common field and at least one user-specific field. The common field can indicate RU distributions to multiple STAs **104**, indicate the RU assignments in the frequency domain, indicate which RUs are allocated for MU-MIMO transmissions and which RUs correspond to MU-OFDMA transmissions, and the number of users in allocations, among other examples. The common field may be encoded with common bits, CRC bits, and tail bits. The user-specific fields are assigned to particular STAs **104** and may be used to schedule specific RUs and to indicate the scheduling to other WLAN devices. Each user-specific field may include multiple user block fields. Each user block field may include, for example, two user fields that contain information for two respective STAs to decode their respective RU payloads.

[0066] The presence of RL-SIG **364** and U-SIG **366** may indicate to EHT-or later version-compliant STAs **104** that the PPDU **350** is an EHT PPDU or a PPDU conforming to any later (post-EHT) version of a new wireless communication protocol conforming to a future IEEE 802.11 wireless communication protocol standard. For example, U-SIG **366** may be used by a receiving device to interpret bits in one or more of EHT-SIG **368** or the data field **374**.

[0067] As described above, APs **102** and STAs **104** can support multi-user (MU) communications; that is, concurrent transmissions from one device to each of multiple devices (for example, multiple simultaneous downlink (DL) communications from an AP **102** to corresponding STAs **104**), or concurrent transmissions from multiple devices to a single device (for example, multiple simultaneous uplink (UL) transmissions from corresponding STAs **104** to an AP **102**). To support the MU transmissions, the APs **102** and STAs **104** may utilize multi-user multiple-input, multiple-output (MU-MIMO) and multi-user orthogonal frequency division multiple access (MU-OFDMA) techniques.

[0068] In MU-OFDMA schemes, the available frequency spectrum of the wireless channel may be divided into multiple resource units (RUs) each including multiple frequency subcarriers (also referred to as “tones”). Different RUs may be allocated or assigned by an AP **102** to different STAs **104** at particular times. The sizes and distributions of the RUs may be referred to as an RU allocation. In some implementations, RUs may be allocated in 2 MHz intervals, and as such, the smallest RU may include 26 tones consisting of 24 data tones and 2 pilot tones. Consequently, in a 20 MHz channel, up to 9 RUs (such as 2 MHz, 26-tone RUs) may be allocated (because some tones are reserved for other purposes). Similarly, in a 160 MHz channel, up to 72 RUs may be allocated. Larger 52 tone, 106 tone, 242 tone, 484 tone and 996 tone RUs may also be allocated. Adjacent RUs may be separated by a null subcarrier (such as a DC subcarrier), for example, to reduce interference between adjacent RUs, to reduce receiver DC offset, and to avoid transmit center frequency leakage.

[0069] For UL MU transmissions, an AP **102** can transmit a trigger frame to initiate and synchronize an UL MU-OFDMA or UL MU-MIMO transmission from multiple STAs **104** to the AP **102**. Such trigger frames may thus enable multiple STAs **104** to send UL traffic to the AP **102** concurrently in time. A trigger frame may address one or more STAs **104** through respective association identifiers (AIDs), and may assign each AID (and thus each STA **104**) one or more RUs that can be used to send UL traffic to the AP **102**. The AP also may designate one or more random access (RA) RUs that unscheduled STAs **104** may contend for.

[0070] Certain aspects of the present disclosure may support allowing APs **104** to allocate STAs

106 transmissions in optimized ways to improve efficiency. Both high efficiency wireless (HEW) stations, stations utilizing an 802.11 high efficiency protocol (such as IEEE 802.11ax), and stations using older or legacy 802.11 protocols (such as 802.11b), can compete or coordinate with each other in accessing a wireless medium.

[0071] In some implementations, APs **104** can transmit on a wireless medium according to various DL tone plans for HEW STAs. For example, with respect to FIG. **1**, the STAs **106A-106D** can be HEW STAs. In some implementations, the HEW STAs can communicate using a symbol duration four times that of a legacy STA. Accordingly, each symbol which is transmitted may be four times as long in duration. When using a longer symbol duration, each of the individual tones may use one-quarter as much bandwidth to be transmitted. For example, in various implementations, a $1\times$ symbol duration can be $3.2\ \mu\text{s}$, a $2\times$ symbol duration can be $6.4\ \mu\text{s}$, and a $4\times$ symbol duration can be $12.8\ \mu\text{s}$. The AP **104** can transmit messages to the HEW STAs **106A-106D** according to one or more tone plans, based on a communication bandwidth. In some aspects, the AP **104** may be configured to transmit to multiple HEW STAs simultaneously, using OFDMA.

[0072] FIG. **4** shows an example 2N-tone plan **400**. In some implementations, the tone plan **400** may correspond to OFDM tones, in the frequency domain, generated using a 2N-point fast Fourier transform (FFT). The tone plan **400** includes 2N OFDM tones indexed-N to N-1. The tone plan **400** includes two sets of edge or guard tones **410**, two sets of data/pilot tones **420**, and a set of direct current (DC) tones **430**. In some implementations, the edge or guard tones **410** and DC tones **430** can be null. In some implementations, the tone plan **400** may include another suitable number of pilot tones or may include pilot tones at other suitable tone locations.

[0073] In some aspects, OFDMA tone plans may be provided for transmission using a $4\times$ symbol duration, as compared to various IEEE 802.11 protocols. For example, $4\times$ symbol duration may use a number of symbols which can be each $12.8\ \mu\text{s}$ in duration (different from symbols in certain other IEEE 802.11 protocols which may be $3.2\ \mu\text{s}$ in duration).

[0074] In some aspects, OFDMA tone plans may be provided for transmission using a $2\times$ symbol duration, as compared to various IEEE 802.11 protocols. For example, the $2\times$ symbol duration may use a number of symbols which can be each $6.4\ \mu\text{s}$ in duration (different from symbols in certain other IEEE 802.11 protocols which may be $3.2\ \mu\text{s}$ or $12.8\ \mu\text{s}$ in duration).

[0075] In some aspects, the data/pilot tones **420** of a transmission **400** may be divided among any number of different users. For example, the data/pilot tones **420** may be divided among between one and eight users. In order to divide the data/pilot tones **420**, an AP **104** or another device may signal to the various devices, indicating which devices may transmit or receive on which tones (of the data/pilot tones **420**) in a particular transmission. Accordingly, systems and methods for dividing the data/pilot tones **420** may be desired, and this division may be based upon a tone plan.

[0076] A tone plan may be chosen based on a number of different characteristics. For example, it may be beneficial to have a simple tone plan, which can be consistent across most or all bandwidths. For example, an OFDMA transmission may be transmitted over 20, 40, 80, 160, 240, or 320 MHz (or a combination thereof), and it may be desirable to use a tone plan that can be used for any of these bandwidths. Further, a tone plan may be simple in that it uses a smaller number of building block sizes. For example, a tone plan may contain a unit which may be referred to as resource unit (RU). This unit may be used to assign a particular amount of wireless resources (for example, bandwidth or particular tones) to a particular user. For example, one user may be assigned bandwidth as a number of RUs, and the data/pilot tones **420** of a transmission may be broken up into a number of RUs.

[0077] A tone plan also may be chosen based on efficiency. For example, transmissions of different bandwidths (for example, 20, 40, 80, 160, 240, or 320 MHz, or a combination thereof) may have different numbers of tones. Reducing the number of leftover tones may be beneficial. Further, it may be beneficial if a tone plan is configured to preserve 20, 40, 80, 160, 240, or 320 MHz boundaries in some implementations. For example, it may be desirable to have a tone plan which

allows each 20, 40, 80, 160, 240, or 320 MHz portion to be decoded separately from each other, rather than having allocations which can be on the boundary between two different 20, 40, 80, 160, 240, or 320 MHz portions of the bandwidth. For example, it may be beneficial for interference patterns to be aligned with 20, 40, 80, 160, 240, or 320 MHz channels. Further, it may be beneficial to have channel bonding, which also may be known as preamble puncturing, such that when a 20 MHz transmission and a 40 MHz transmission can be transmitted, to create a 20 MHz “hole” in the transmission when transmitted over 80, 160, 240, or 320 MHz. This may allow, for example, a legacy packet to be transmitted in this unused portion of the bandwidth. This puncturing may apply to any transmission (for example, 20, 40, 80, 160, 240, or 320 MHz transmissions) and may create “holes” of at least 20 MHz in the transmission regardless of the channel or bandwidth being used. Finally, it also may be advantageous to use a tone plan which provides for fixed pilot tone locations in various transmissions, such as in different bandwidths.

[0078] As data transmission rate demands increase with additional devices joining networks or additional data being added for transmission over networks, larger channel bandwidths may be introduced, for example for orthogonal frequency-division multiple access (OFDMA) transmissions. In one example, tone plans for a 320 MHz total channel bandwidth may be introduced to assist in increasing peak system transmission data rates and to more efficiently utilize available channels. For example, as new frequencies are available for use (for example, 6 GHz), these new tone plans for the larger total channel bandwidths may more efficiently utilize the newly available channels. Moreover, an increased total bandwidth which may be provided by these new tone plans may allow for better rate vs range tradeoff. In this case, the same or a similar transmission rate may be used to provide larger coverage if a larger total bandwidth is used. Additionally, the larger total channel bandwidths also may increase tone plan efficiency (for example, for a particular BW, how many tones could be used for data transmission) and also may increase a number of guard bands. As with any total channel bandwidth being used, different modes may be available depending on channel availability. For example, current 80 MHz channel bandwidths may be separated into 20 MHz, 40 MHz, or 80 MHz modes.

[0079] FIG. 5 shows example modes **500a-500d** available for 320 MHz bandwidth transmissions and 240 MHz bandwidth transmissions. These may be some of the bandwidth modes available in IEEE 802.11be. As shown in FIG. 5, the 320 MHz bandwidth transmissions may be transmitted in at least two different modes as shown in **500a** and **500b**. Each of the modes **500a** and **500b** may represent a different combination of channel bandwidth (BW) and frequency bands that may be used, depending on channel availability (for example, in 2.4, 5, or 6 GHz networks). In a first mode **500a**, the 320 MHz transmission may be transmitted in a single, contiguous frequency band having 320 MHz bandwidth. In a second mode **500b**, the 320 MHz transmission may be transmitted in two non-contiguous, disjoint frequency bands having 160 MHz bandwidth and 80 MHz bandwidth, respectively. As shown, each of the frequency bands is separated by unused sub-bands (SBs). In this context, unused SBs refer to portions of the frequency band that are not part of the wireless channel.

[0080] Similarly, the 240 MHz bandwidth transmissions may be transmitted in at least two different modes as shown in **500c** and **500d**. In a third mode **500c**, the 240 MHz transmission may be transmitted in a single, contiguous frequency band having 240 MHz bandwidth. In a fourth mode **500d**, the 240 MHz transmission may be transmitted in two non-contiguous, disjoint frequency bands having 160 MHz bandwidth and 80 MHz bandwidth, respectively. As shown, each of the frequency bands is separated by unused sub-bands (SBs). In this context, unused SBs refer to portions of the frequency band that are not part of the wireless channel.

[0081] In some implementations, tone plans may be designed and signal generation may be completed for contiguous frequency band of 80 MHz, 160 MHz, and 320 MHz bandwidth sizes.

[0082] Each of the modes **500a-500d** may have one or more options for creating the 320 MHz bandwidth transmission or the 240 MHz bandwidth transmission. The first mode **500a** may include

(1) a first option of having a single 320 MHz tone plan; (2) a second option of duplicating two 160 MHz tone plans, one in each of two PHY 160 MHz subchannels and separated by an unused SB; and (3) a third option of duplicating four 80 MHz tone plans, one in each of four PHY 80 MHz subchannels separated by unused SBs.

[0083] The second mode **500b** may include (1) a first option of using two 160 MHz tone plans, each in one PHY 160 MHz subchannel and (2) a second option of duplicating four 80 MHz tone plans, one in each of four PHY 80 MHz subchannels and separated by unused SBs. The third mode **500c** may include (1) a first option of having a single 240 MHz tone plan; (2) a second option of one 160 MHz tone plan in one PHY 160 MHz subchannel and one 80 MHz subchannel in one PHY 80 MHz subchannel and separated by an unused SB; and (3) a third option of duplicating three 80 MHz tone plans, one in each of three PHY 80 MHz subchannels separated by unused SBs.

[0084] Based on these modes and options, different tone plans may be designed or generated for the 80, 160, or 320 MHz bandwidths. Tone plan designs for 80 MHz, 160 MHz, and 320 MHz for 3 symbol duration options are the building blocks. In some implementations, the different frequency bands may use different symbol durations. For example, for the third option of the 320 MHz frequency band, the 160 MHz frequency band may use a first symbol duration while the 80 MHz frequency bands may use a second symbol duration different from the first symbol duration. In some implementations, the tone plans for the 320 MHz bandwidth may be generated or designed based on the building blocks (for example, the 80 and 160 MHz transmissions discussed herein).

[0085] The different modes described herein for the 320 MHz channel bandwidth and 240 MHz channel bandwidth may provide different options of symbol durations and tone spacings, depending on the mode being used.

[0086] FIG. 6 illustrates examples of OFDMA resource unit (RU) allocations. In an OFDMA system **600**, multiple client devices (stations) may communicate (transmit or receive) with an access point (AP) at the same time by sharing available bandwidth. OFDMA allows sub-carriers (also referred to as “tones”) in a channel bandwidth to be grouped into smaller portions called “Resource Units” (RU). Each RU may consist of a group of tones. In various example implementations, an RU may consist of 26 tones, 52 tones, 106 tones, 242 tones, 484 tones, or 996 tones. That is, RUs may have different sizes depending on the number of tones or subcarriers in an RU, such that for example an RU₂₆ contains 26 tones and an RU₅₂ contains 52 tones, and so on. These individual RU's are assigned to different client devices or stations, which allows the access point to serve them simultaneously during uplink and downlink transmissions. In the OFDMA allocation system **600**, individual RUs may be allocated to each device communicating over the wireless network, the RUs need not be contiguous.

[0087] Although not illustrated in FIG. 6, some RUs may be scheduled to carry one or more physical channels, including control channels, shared channels, data channels, among other examples. Other RUs may also carry pilots or reference signals. These pilots or reference signals (for example, including long training fields or LTFs) may provide for a receiving device to perform channel estimation of the corresponding channel, which may enable coherent demodulation/detection of the control or data channels within the RU.

[0088] FIG. 7 illustrates examples of tone spacings and index ranges for various fast Fourier transform (FFT) sizes and symbol durations for 80 MHz, 160 MHz, and 320 MHz transmissions. Various 802.11 protocols may use 1× symbol durations. The 1× symbol durations may have a tone spacing of 312.5 kHz. Some 802.11 protocols may also use 4× symbol durations. The 4× symbol durations may have a tone spacing of 78.125 kHz. Next generation 802.11 devices and standards may utilize either the 1× or 4× symbol durations and may also utilize a 2× symbol duration of 6.4 μs having a tone spacing of 156.25 kHz. Specifically, FIG. 7 shows an FFT size for each option (for example, combination of symbol duration and tone spacing). For example, the 80 MHz channel bandwidth (BW) has 256 tones available at 1× symbol duration and 312.5 kHz spacing (option 1), 512 tones available at 2× symbol duration and 156.25 kHz spacing (option 2), and 1024

tones available at $4\times$ symbol duration and 78.125 kHz spacing (option 3). The 160 MHz channel BW has 512 tones available at $1\times$ symbol duration and 312.5 kHz spacing, 1024 tones available at $2\times$ symbol duration and 156.25 kHz spacing, and 2048 tones available at $4\times$ symbol duration and 78.125 kHz spacing. The 320 MHz channel BW has 1024 tones available at $1\times$ symbol duration and 312.5 kHz spacing, 2048 tones available at $2\times$ symbol duration and 156.25 kHz spacing, and 4096 tones available at $4\times$ symbol duration and 78.125 kHz spacing. In some aspects, $1\times$ and $2\times$ symbol durations may have similar benefits as compared to a $4\times$ symbol durations. In some aspects, $1\times$ and $2\times$ symbol durations may have lower complexity, latency, and memory requirements due to corresponding smaller Fast Fourier Transform (FFT) sizes as compared to the $4\times$ symbol duration, which has a higher complexity, latency, and memory requirement due to its larger FFT size. The $1\times$ and $2\times$ symbol durations each have a lower tone plan and cyclic prefix (CP) or guard interval (GI) efficiency than the $4\times$ symbol duration that has a higher tone plan and GI efficiency. Furthermore, the $1\times$ and $2\times$ symbol durations may not have outdoor support while the $4\times$ symbol duration may have outdoor support, although the 320 MHz bandwidth may be generally used indoors. The $1\times$ and $2\times$ symbol durations may need a new design to provide OFDMA support, as they cannot be mixed with high efficiency STAs in DL/UL OFDMA. However, the $4\times$ symbol duration may provide OFDMA support, as it can be mixed with HE STAs in DL/UL OFDMA. When memory size is not being considered, then the $4\times$ symbol duration may be a more natural choice for symbol duration. However, if maintaining memory size is the goal, then the $1\times$ or $2\times$ symbol duration may be considered. For $1\times$ trigger based PPDU, the UL overhead of 50% in view of the $1.6\ \mu\text{s}$ GI is too high, so the $2\times$ symbol duration may be more likely a selection. In some implementations, reduced symbol durations may advantageously result in reduced complexity and reduced memory utilization.

[0089] In one example, data symbols may have a $4\times$ symbol duration, which is $12.8\ \mu\text{s} + \text{GI}$ (guard interval). By contrast, LTF symbols may have various multiples of a symbol duration. A $1\times$ LTF uses $1\times$ symbol duration, which is $3.2\ \mu\text{s} + \text{GI}$. A $2\times$ LTF uses $2\times$ symbol duration, which is $6.4\ \mu\text{s} + \text{GI}$. A $4\times$ LTF uses $4\times$ symbol duration, which is $12.8\ \mu\text{s} + \text{GI}$, same as data symbols. In 802.11ax, defines three different GI values: $0.8\ \mu\text{s}$, $1.6\ \mu\text{s}$, and $3.2\ \mu\text{s}$. Each one of the three values may be selected based on the PPDU format and channel condition. The same GI may be applied to LTF field and data field. For packets using $1\times$ LTF, either $0.8\ \mu\text{s}$ or $1.6\ \mu\text{s}$ GIs can be used depending on the PPDU format. For packets using $2\times$ LTF, either $0.8\ \mu\text{s}$ or $1.6\ \mu\text{s}$ GIs can be used depending on the PPDU format. For packets using $4\times$ LTF, either $0.8\ \mu\text{s}$, or $3.2\ \mu\text{s}$ GIs may be used depending on PPDU format or channel condition.

[0090] Accordingly, the index ranges of the tones for each of these options is shown in FIG. 7, which shows that 256 tones have a range of $[-128, 127]$, 512 tones have a range of $[-256, 255]$, 1024 tones have a range of $[-512, 511]$, 2048 tones have a range of $[-1024, 1023]$, and 4096 tones have a range from $[-2048, 2047]$.

[0091] As the IEEE 802.11 standards evolve to use higher bandwidths (for example, 320 MHz in IEEE 802.11be), it has been recognized that OFDM transmissions can have high peak values in the time domain since many subcarrier components are added via an inverse fast Fourier transformation (IFFT) operation. Consequently, OFDM transmissions may have a high peak-to-average power ratio (PAPR) when compared to single-carrier transmissions. The high PAPR of OFDM and OFDMA transmissions decreases the signal-to-quantization noise ratio (SQNR) of the analog-digital convertor (ADC) and digital-analog convertor (DAC) in a radio chain while degrading the efficiency of the power amplifier in a transmitter.

[0092] A long training field (LTF) is used for wireless communications transmissions, including over a large bandwidth channel such as a 320 MHz bandwidth channel used for IEEE 802.11be. The LTF includes a training sequence that can be used by a receiving device to set or adjust an automatic gain control (AGC) function of the receiving device. AGC is a technique in an amplifier or chain of amplifiers used to adjust the signal amplitude at its output regardless of the variation of

the signal amplitude at the input. The LTF may include a training sequence of predetermined signals that can be used to identify the beginning of the transmission and to set the AGC.

[0093] The LTF sequence may serve as a reference signal that enables a receiving device to perform channel estimation of the corresponding channel, which may enable coherent demodulation/detection of the control or data channels.

[0094] As 802.11be seeks extreme high throughput (EHT) wireless communications using large bandwidth channels and OFDM modulation, it would be advantageous to configure the LTF to minimize or reduce the peak-to-average-power ratio (PAPR) of the LTF so that the LTF does not skew the AGC incorrectly.

[0095] One solution is to reutilize a High Efficiency (HE)-LTF sequence defined by the IEEE 802.11 standards for 80 MHz bandwidth channels as a basis to construct a longer EHT-LTF sequence for 320 MHz bandwidth channels. In order to minimize the PAPR of the EHT-LTF for 320 MHz bandwidth channels, the HE-LTF for 80 MHz bandwidth channels may be modified by phase rotations and concatenations to achieve a minimized PAPR or a PAPR below a threshold level. One way to achieve this may be to conduct a simulation to search for a combination of phase rotation coefficients and HE-LTF segments that provide such minimized PAPR.

[0096] FIG. 8 illustrates an example of an Extreme High Throughput (EHT) long training field (LTF) (EHT-LTF) construction for a 320 MHz bandwidth channel, where each symbol of the EHT-LTF uses a $1\times$ symbol duration and each symbol of a data field uses a $4\times$ symbol duration. For example, each symbol of the EHT-LTF sequence may have a symbol duration of $3.2\text{ }\mu\text{s}$ plus a guard interval (GI) while each symbol of the data field may use $12.8\text{ }\mu\text{s} + \text{GI}$ (guard interval).

[0097] The EHT-LTF sequence for the 320 MHz bandwidth channel is based on a concatenation of four LTF subsequences, each covering an 80 MHz bandwidth channel. In this example, the four LTF subsequences are defined (LTF.sub.80MHz_lower1_1x, LTF.sub.80MHz_upper1_1x, LTF.sub.80MHz_lower2_1x, LTF.sub.80MHz_upper2_1x) with filler zeros (for example, 23 zeros) in between. Each of the LTF subsequences may be a function of different portions, segments, or subsequences of the HE-LTF sequence for an 80 MHz bandwidth channel that is selectively phase-rotated by a set of phase rotation values to minimize PAPR for the 320 MHz bandwidth channel or a 240 MHz bandwidth channel. FIG. 9 illustrates examples of High Efficiency (HE) LTF (HE-LTF) subsequences for an 80 MHz bandwidth channel including a first LTF base sequence and a second LTF base sequence. These HE-LTF subsequences are specified in the IEEE 802.11ax standard.

[0098] A first LTF subsequence (LTF.sub.80MHz_lower1_1x) may be a combination of the first LTF base sequence LTF-A that is phase-rotated by a first rotation value s_1 , concatenated with the second LTF base sequence LTF-B that is phase-rotated by a second rotation value s_2 , and a zero in between both sequences.

[0099] A second LTF subsequence (LTF.sub.80MHz_upper1_1x) may be a combination of the first LTF base sequence LTF-A that is phase-rotated by a third rotation value s_3 , concatenated with the second LTF base sequence LTF-B that is phase-rotated by a fourth rotation value s_4 , and a zero in between both sequences.

[0100] A third LTF subsequence (LTF.sub.80MHz_lower2_1x) may be a combination of the first LTF base sequence LTF-A that is phase-rotated by a fifth rotation value s_5 , concatenated with the second LTF base sequence LTF-B that is phase-rotated by a sixth rotation value s_6 , and a zero in between both sequences.

[0101] A fourth LTF subsequence (LTF.sub.80MHz_upper2_1x) may be a combination of the first LTF base sequence LTF-A that is phase-rotated by a seventh rotation value s_7 , concatenated with the second LTF base sequence LTF-B that is phase-rotated by an eighth rotation value s_8 , and a zero in between both sequences.

[0102] In one implementation, the phase rotation values $s=[s_1\ s_2\ s_3\ s_4\ s_5\ s_6\ s_7\ s_8]$ may be $[1\ 1\ 1\ 1\ -1\ -1\ -1\ -1]$ to minimize or reduce PAPR for the EHT-LTF for a 320 MHz bandwidth channel or a 240 MHz bandwidth channel.

[0103] In some implementations, the phase rotation values or coefficients $s=[s_1 s_2 s_3 s_4 s_5 s_6 s_7 s_8]$ may be scaled or multiplied by a common scaling factor b without affecting the optimization to minimize the PAPRs. Additionally, the various LTF subsequences may be concatenated in different orders without departing from the present disclosure. For instance, the EHT-LTF sequence may be the concatenation of: second LTF subsequence, k zeros, third LTF subsequence, k zeros, fourth LTF subsequence, k zeros, and first LTF subsequence. In another example, the EHT-LTF sequence may be the concatenation of: fourth LTF subsequence, k zeros, second LTF subsequence, k zeros, first LTF subsequence, k zeros, and third LTF subsequence. In summary, any combination or order of the LTF subsequences may be used to derive the EHT-LTF sequence.

[0104] FIG. 10 illustrates an example of an EHT-LTF construction for a 320 MHz bandwidth channel, where each symbol of the EHT-LTF uses a $2\times$ symbol duration and each symbol of a data field uses a $4\times$ symbol duration. For example, each symbol of the EHT-LTF sequence may have a symbol duration of $6.4\text{ }\mu\text{s}$ plus a guard interval (GI) while each symbol of the data field may use $12.8\text{ }\mu\text{s}+\text{GI}$ (guard interval).

[0105] The EHT-LTF sequence for the 320 MHz bandwidth channel is based on a concatenation of four LTF subsequences, each covering an 80 MHz bandwidth channel. In this example, the four LTF subsequences are defined (LTF.sub.80MHz_lower1_2x, LTF.sub.80MHz_upper1_2x, LTF.sub.80MHz_lower2_2x, LTF.sub.80MHz_upper2_2x) with filler zeros (for example, 23 zeros) in between. Each of the LTF subsequences may be a function of different portions, segments, subsequences of the HE-LTF sequence for an 80 MHz bandwidth channel that is selectively phase-rotated by a set of rotation values to minimize PAPR for the 320 MHz bandwidth channel or a 240 MHz bandwidth channel. FIG. 11 illustrates examples of HE-LTF subsequences for an 80 MHz bandwidth channel including a first LTF base sequence LTF-A, a second LTF base sequence LTF-B, a third LTF base sequence LTF-C, a fourth LTF base sequence LTF-D, and a fifth LTF base sequence LTF-E. These HE-LTF subsequences are specified in the IEEE 802.11ax standard.

[0106] A first LTF subsequence (LTF.sub.80MHz_lower1_2x) may be a combination of the first LTF base sequence LTF-A that is phase-rotated by a first rotation value q_1 , concatenated with the second LTF base sequence LTF-B that is phase-rotated by a second rotation value q_2 , concatenated with the third LTF base sequence LTF-C that is phase-rotated by a third rotation value q_3 , concatenated with the fourth LTF base sequence LTF-D that is phase-rotated by a fourth rotation value q_4 , and concatenated with the fifth LTF base sequence LTF-E that is phase-rotated by a fifth rotation value q_5 .

[0107] A second LTF subsequence (LTF.sub.80MHz_upper1_2x) may be a combination of the first LTF base sequence LTF-A that is phase-rotated by a sixth rotation value q_6 , concatenated with the second LTF base sequence LTF-B that is phase-rotated by a seventh rotation value q_7 , concatenated with the third LTF base sequence LTF-C that is phase-rotated by an eighth rotation value q_8 , concatenated with the fourth LTF base sequence LTF-D that is phase-rotated by a ninth rotation value q_9 , and concatenated with the fifth LTF base sequence LTF-E that is phase-rotated by a tenth rotation value q_{10} .

[0108] A third LTF subsequence (LTF.sub.80MHz_lower2_2x) may be a combination of the first LTF base sequence LTF-A that is phase-rotated by an eleventh rotation value q_{11} , concatenated with the second LTF base sequence LTF-B that is phase-rotated by a twelfth rotation value q_{12} , concatenated with the third LTF base sequence LTF-C that is phase-rotated by a thirteenth rotation value q_{13} , concatenated with the fourth LTF base sequence LTF-D that is phase-rotated by a fourteenth rotation value q_{14} , and concatenated with the fifth LTF base sequence LTF-E that is phase-rotated by a fifteenth rotation value q_{15} .

[0109] A fourth LTF subsequence (LTF.sub.80MHz_upper2_2x) may be a combination of the first LTF base sequence LTF-A that is phase-rotated by a sixteenth rotation value q_{16} , concatenated with the second LTF base sequence LTF-B that is phase-rotated by a seventeenth rotation value q_{17} , concatenated with the third LTF base sequence LTF-C that is phase-rotated by an eighteenth

rotation value q18, concatenated with the fourth LTF base sequence LTF-D that is phase-rotated by a nineteenth rotation value q19, and concatenated with the fifth LTF base sequence LTF-E that is phase-rotated by a twentieth rotation value q20.

[0110] In one implementation, the phase rotation values $q=[q_1\ q_2\ q_3\ q_4\ q_5\ q_6\ q_7\ q_8\ q_9\ q_{10}\ q_{11}\ q_{12}\ q_{13}\ q_{14}\ q_{15}\ q_{16}\ q_{17}\ q_{18}\ q_{19}\ q_{20}]$ may be $[+1\ -1\ -1\ -1\ +1\ -1\ +1\ +1\ +1\ -1\ -1\ +1\ +1\ +1\ -1\ -1\ +1\ -1\ -1\ +1]$ or $[+1\ +1\ -1\ -1\ -1\ -1\ -1\ -1\ -1\ -1\ +1\ +1\ +1\ +1\ +1\ +1\ +1\ +1\ +1]$ to minimize PAPR for the EHT-LTF for a 320 MHz bandwidth channel or a 240 MHz bandwidth channel.

[0111] In some implementations, the phase rotation values or coefficients $q=[q_1\ q_2\ q_3\ q_4\ q_5\ q_6\ q_7\ q_8\ q_9\ q_{10}\ q_{11}\ q_{12}\ q_{13}\ q_{14}\ q_{15}\ q_{16}\ q_{17}\ q_{18}\ q_{19}\ q_{20}]$ may be scaled or multiplied by a common scaling factor b without affecting the optimization to minimize the PAPRs. Additionally, the various LTF subsequences may be concatenated in different orders without departing from the present disclosure. For instance, the EHT-LTF sequence may be the concatenation of: second LTF subsequence, k zeros, fourth LTF subsequence, k zeros, third LTF subsequence, k zeros, and first LTF subsequence. In another example, the EHT-LTF sequence may be the concatenation of: fourth LTF subsequence, k zeros, first LTF subsequence, k zeros, third LTF subsequence, k zeros, and second LTF subsequence. In summary, any combination or order of the LTF subsequences may be used to derive the EHT-LTF sequence.

[0112] FIG. 12 illustrates another example of an EHT-LTF construction for a 320 MHz bandwidth channel, where each symbol of the EHT-LTF uses a $2\times$ symbol duration and each symbol of a data field uses a $4\times$ symbol duration. For example, each symbol of the EHT-LTF sequence may have a symbol duration of $6.4\ \mu\text{s}$ plus a guard interval (GI) while each symbol of the data field may use $12.8\ \mu\text{s}+\text{GI}$ (guard interval).

[0113] The EHT-LTF sequence for the 320 MHz bandwidth channel is based on a concatenation of four LTF subsequences, each covering an 80 MHz bandwidth channel. In this example, the four LTF subsequences are defined (LTF.sub.80MHz_lower1_2x, LTF.sub.80MHz_upper1_2x, LTF.sub.80MHz_lower2_2x, LTF.sub.80MHz_upper2_2x) with filler zeros (for example, 23 zeros) in between. Each of the LTF subsequences may be a function of different portions, segments, subsequences of the HE-LTF sequence for an 80 MHz bandwidth channel that is selectively phase-rotated by a set of rotation values to minimize PAPR for the 320 MHz bandwidth channel or a 240 MHz bandwidth channel. FIG. 13 illustrates examples of HE-LTF subsequences for an 80 MHz bandwidth channel including a first LTF base sequence LTF-A and a second LTF base sequence LTF-B.

[0114] In one example, the HE-LTF subsequences (the first LTF base sequence LTF-A and the second LTF base sequence LTF-B) may be based on an 80 MHz LTF segment formed from two High Efficiency (HE) LTF (HE-LTF) sequences for subchannels having a 40 MHz channel bandwidth. The 40 MHz channel bandwidth sequences are modified to add optimized values for missing tones used in the 80 MHz channel bandwidth. Once the 80 MHz LTF segment is obtained, it may be split to obtain the first LTF base sequence LTF-A and the second LTF base sequence LTF-B.

[0115] For instance, the subsequences for the 80 MHz bandwidth channel may be formed by interleaving and phase rotating values of four base sub-segments S_a , S_b , S_c , and S_d , where [0116] $S_a=[+1\ +1\ +1\ +1\ +1\ -1\ -1\ +1\ +1\ -1\ +1\ -1\ +1]$, [0117] $S_b=[S_a(1:7), -S_a(8:13)]$, [0118] $S_c=[-S_a(1:6), S_a(7:13)]$, and [0119] $S_d=[S_d(1:14)]$, where the values $S_d(x)$ are configured to reduce a peak-to-average power ratio in an 80 MHz segment. In one example, $S_d=[-1, -1, -1, -1, 1, -1, -1, -1, 1, 1, 1, -1, -1, -1]$. The sub-sequences S_a , S_b , and S_c may be defined for HE-LTF, and the sub-sequence S_d may be optimized to add tones and reduce PAPR.

[0120] As illustrated in FIG. 12, a first LTF subsequence (LTF.sub.80MHz_lower1_2x) may be a combination of the first LTF base sequence LTF-A that is phase-rotated by a first rotation value q_1 , concatenated with the second LTF base sequence LTF-B that is phase-rotated by a second rotation value q_2 .

[0121] A second LTF subsequence (LTF.sub.80MHz_upper1_2x) may be a combination of the first LTF base sequence LTF-A that is phase-rotated by a third rotation value q_3 , concatenated with the second LTF base sequence LTF-B that is phase-rotated by a fourth rotation value q_4 .

[0122] A third LTF subsequence (LTF.sub.80MHz_lower2_2x) may be a combination of the first LTF base sequence LTF-A that is phase-rotated by a fifth rotation value q_5 , concatenated with the second LTF base sequence LTF-B that is phase-rotated by a sixth rotation value q_6 .

[0123] A fourth LTF subsequence (LTF.sub.80MHz_upper2_2x) may be a combination of the first LTF base sequence LTF-A that is phase-rotated by a seventh rotation value q_7 , concatenated with the second LTF base sequence LTF-B that is phase-rotated by an eighth rotation value q_8 .

[0124] In one implementation, the phase rotation values $q=[q_1 \ q_2 \ q_3 \ q_4 \ q_5 \ q_6 \ q_7 \ q_8]$ may be $[1 \ 1 \ 1 \ 1 \ -1 \ -1 \ -1 \ 1]$ to minimize PAPR for the EHT-LTF for a 320 MHz bandwidth channel or a 240 MHz bandwidth channel.

[0125] In some implementations, the phase rotation values or coefficients $q=[q_1 \ q_2 \ q_3 \ q_4 \ q_5 \ q_6 \ q_7 \ q_8]$ may be scaled or multiplied by a common scaling factor b without affecting the optimization to minimize the PAPRs. Additionally, the various LTF subsequences may be concatenated in different orders without departing from the present disclosure. For instance, the EHT-LTF sequence may be the concatenation of: first LTF subsequence, k zeros, fourth LTF subsequence, k zeros, third LTF subsequence, k zeros, and second LTF subsequence. In another example, the EHT-LTF sequence may be the concatenation of: second LTF subsequence, k zeros, fourth LTF subsequence, k zeros, third LTF subsequence, k zeros, and first LTF subsequence. In summary, any combination or order of the LTF subsequences may be used to derive the EHT-LTF sequence.

[0126] FIG. 14 illustrates a first example of an EHT long training field (LTF) construction for a 320 MHz bandwidth channel, where each symbol of the EHT-LTF uses a $4\times$ symbol duration and each symbol of a data field uses a $4\times$ symbol duration. For example, each symbol of the EHT-LTF sequence may have a symbol duration of $12.8 \mu\text{s}$ plus a guard interval (GI) while each symbol of the data field may use $12.8 \mu\text{s}$ plus GI (guard interval).

[0127] The EHT-LTF sequence for the 320 MHz bandwidth channel is based on a concatenation of four subsequences, each covering an 80 MHz bandwidth channel. In this example, the four LTF subsequences are defined (LTF.sub.80MHz_lower1_4x, LTF.sub.80MHz_upper1_4x, LTF.sub.80MHz_lower2_4x, LTF.sub.80MHz_upper2_4x) with filler zeros (for example, 23 zeros) in between. Each of the LTF subsequences may be a function of different portions, segments, or subsequences of the HE-LTF sequence for an 80 MHz bandwidth channel that is selectively phase-rotated by a set of phase rotation values to minimize PAPR for the 320 MHz bandwidth channel or a 240 MHz bandwidth channel. FIG. 15 illustrates examples of HE-LTF subsequences for an 80 MHz bandwidth channel including a first LTF base sequence LTF-A, a second LTF base sequence LTF-B, a third LTF base sequence LTF-C, and a fourth LTF base sequence LTF-D.

[0128] A first LTF subsequence (LTF.sub.80MHz_lower1_4x) may be a combination of a first LTF base sequence LTF-A that is phase-rotated by a first rotation value n_1 , concatenated with a second LTF base sequence LTF-B that is phase-rotated by a second rotation value n_2 , followed by a zero, concatenated with the third LTF base sequence LTF-C that is phase-rotated by a third rotation value n_3 , concatenated with the fourth LTF base sequence LTF-D that is phase-rotated by a fourth rotation value n_4 .

[0129] A second LTF subsequence (LTF.sub.80MHz_upper1_4x) may be a combination of the first LTF base sequence LTF-A that is phase-rotated by a fifth rotation value n_5 , concatenated with the second LTF base sequence LTF-B that is phase-rotated by a sixth rotation value n_6 , followed by a zero, concatenated with the third LTF base sequence LTF-C that is phase-rotated by a seventh rotation value n_7 , concatenated with the fourth LTF base sequence LTF-D that is phase-rotated by an eighth rotation value n_8 .

[0130] A third LTF subsequence (LTF.sub.80MHz_lower2_4x) may be a combination of the first LTF base sequence LTF-A that is phase-rotated by a ninth rotation value n_9 , concatenated with the

second LTF base sequence LTF-B that is phase-rotated by a tenth rotation value n_{10} , followed by a zero, concatenated with the third LTF base sequence LTF-C that is phase-rotated by an eleventh rotation value n_{11} , concatenated with the fourth LTF base sequence LTF-D that is phase-rotated by a twelfth rotation value n_{12} .

[0131] A fourth LTF subsequence (LTF.sub.80MHz_upper2_4x) may be a combination of the first LTF base sequence LTF-A that is phase-rotated by a thirteenth rotation value n_{13} , concatenated with the second LTF base sequence LTF-B that is phase-rotated by a fourteenth rotation value n_{14} , followed by a zero, concatenated with the third LTF base sequence LTF-C that is phase-rotated by a fifteenth rotation value n_{15} , concatenated with the fourth LTF base sequence LTF-D that is phase-rotated by a sixteenth rotation value n_{16} .

[0132] In one implementation, the phase rotation values $n=[n_1 n_2 n_3 n_4 n_5 n_6 n_7 n_8 n_9 n_{10} n_{11} n_{12} n_{13} n_{14} n_{15} n_{16}]$ may be $[1 -1 1 -1 -1 -1 1 1 1 -1 -1 1 1 1 1]$ or $[1 -1 1 1 -1 1 -1 1 1 1 1 1 1 -1 -1]$ to minimize PAPR for the EHT-LTF for a 320 MHz bandwidth channel or a 240 MHz bandwidth channel.

[0133] In some implementations, the phase rotation values or coefficients $n=[n_1 n_2 n_3 n_4 n_5 n_6 n_7 n_8 n_9 n_{10} n_{11} n_{12} n_{13} n_{14} n_{15} n_{16}]$ may be scaled or multiplied by a common scaling factor b without affecting the optimization to minimize the PAPRs. Additionally, the various LTF subsequences may be concatenated in different orders without departing from the present disclosure. For instance, the EHT-LTF sequence may be the concatenation of: third LTF subsequence, k zeros, second LTF subsequence, k zeros, third LTF subsequence, k zeros, and first LTF subsequence. In another example, the EHT-LTF sequence may be the concatenation of: fourth LTF subsequence, k zeros, first LTF subsequence, k zeros, third LTF subsequence, k zeros, and second LTF subsequence. In summary, any combination or order of the LTF subsequences may be used to derive the EHT-LTF sequence.

[0134] FIG. 16 illustrates a second example of an EHT long training field (LTF) construction for a 320 MHz bandwidth channel, where each symbol of the EHT-LTF uses a $4\times$ symbol duration and each symbol of a data field uses a $4\times$ symbol duration. For example, each symbol of the EHT-LTF sequence may have a symbol duration of $12.8\ \mu\text{s}$ plus a guard interval (GI) while each symbol of the data field may use $12.8\ \mu\text{s}$ plus GI (guard interval).

[0135] The EHT-LTF sequence for the 320 MHz bandwidth channel is based on a concatenation of four LTF subsequences, each covering an 80 MHz bandwidth channel. In this example, the four LTF subsequences are defined (LTF.sub.80MHz_lower1_4x, LTF.sub.80MHz_upper1_4x, LTF.sub.80MHz_lower2_4x, LTF.sub.80MHz_upper2_4x) with filler zeros (for example, 23 zeros) in between. Each of the LTF subsequences may be a function of different portions, segments, or subsequences of the HE-LTF sequence for an 80 MHz bandwidth channel that is selectively phase-rotated by a set of phase rotation values to minimize PAPR for the 320 MHz bandwidth channel or a 240 MHz bandwidth channel. FIG. 17 illustrates examples of HE-LTF subsequences for an 80 MHz bandwidth channel including a first LTF base sequence and a second LTF base sequence.

[0136] A first LTF subsequence (LTF.sub.80MHz_lower1_4x) may be a combination of the first LTF base sequence LTF-A that is phase-rotated by a first rotation value n_1 , concatenated with the second LTF base sequence LTF-B that is phase-rotated by a second rotation value n_2 , and a zero in between both sequences.

[0137] A second LTF subsequence (LTF.sub.80MHz_upper1_4x) may be a combination of the first LTF base sequence LTF-A that is phase-rotated by a third rotation value n_3 , concatenated with the second LTF base sequence LTF-B that is phase-rotated by a fourth rotation value n_4 , and a zero in between both sequences.

[0138] A third LTF subsequence (LTF.sub.80MHz_lower2_4x) may be a combination of the first LTF base sequence LTF-A that is phase-rotated by a fifth rotation value n_5 , concatenated with the second LTF base sequence LTF-B that is phase-rotated by a sixth rotation value n_6 , and a zero in between both sequences.

[0139] A fourth LTF subsequence (LTF.sub.80MHz_upper2_4x) may be a combination of the first LTF base sequence LTF-A that is phase-rotated by a seventh rotation value n_7 , concatenated with the second LTF base sequence LTF-B that is phase-rotated by an eighth rotation value n_8 , and a zero in between both sequences.

[0140] In one implementation, the phase rotation values $n=[n_1\ n_2\ n_3\ n_4\ n_5\ n_6\ n_7\ n_8]$ may be $[1\ -1\ -1\ -1\ 1\ 1\ 1\ 1]$ to minimize PAPR for the EHT-LTF for a 320 MHz bandwidth channel or a 240 MHz bandwidth channel.

[0141] In some implementations, the phase rotation values or coefficients $n=[n_1\ n_2\ n_3\ n_4\ n_5\ n_6\ n_7\ n_8]$ may be scaled or multiplied by a common scaling factor b without affecting the optimization to minimize the PAPRs. Additionally, the various LTF subsequences may be concatenated in different orders without departing from the present disclosure. For instance, the EHT-LTF sequence may be the concatenation of: second LTF subsequence, k zeros, first LTF subsequence, k zeros, fourth LTF subsequence, k zeros, and third LTF subsequence. In another example, the EHT-LTF sequence may be the concatenation of: fourth LTF subsequence, k zeros, second LTF subsequence, k zeros, first LTF subsequence, k zeros, and third LTF subsequence. In summary, any combination or order of the LTF subsequences may be used to derive the EHT-LTF sequence.

[0142] Another aspect provides for the EHT-LTF sequence to be optimized based on multi-stream PAPR performances for different resource units (RU) or multi-RUs (MRUs) (that is combining more than one RU to form a new unit), and depending on the RU size, different PAPR thresholds may be used for optimization. That is, the LTF sequence may be modified to achieve minimized PAPRs or PAPRs below threshold levels based on different (M) RUs.

[0143] In order to derive optimized phase rotation coefficients s , q , or n for the EHT-LTF sequences of FIG. 8, 10, 12, 14, or 16, an exhaustive search may be conducted for these optimized coefficients: s , q , or n , based on the PAPR results for the evaluated RUs or multi-RUs (MRUs). For instance, in the example of a 320 MHz bandwidth channel, RUs and MRU of size: 242, 484, 242+484, 996, 484+996 in a continuous 160 MHz, 2×996 , 3×996 , $3\times 996+484$, 4×996 may be selected for optimization. For each RU or MRU, the multi-stream PAPR may be evaluated when single stream pilot (SSP) is used. For each sequence, the set of coefficients s , q , or n is selected that minimize the PAPR of all above-listed (M) RUs.

[0144] In selecting a set of phase rotation coefficients s , q , or n , it should be observed that for a single stream pilot (SSP), any P value unequal to R value changes the base EHT-LTF sequence. The P matrix and R matrix are rotation matrices applied to data subcarriers and pilot subcarriers when generating a multi-stream symbol, respectively. Thus, different P and R values lead to different PAPRs. EHT-LTF sequence optimization may be done by finding a sequence that minimize the maximal PAPR over all possible P and R values, that is:

[00001]
$$\text{EHT-LTF} = \min_x \{ \max_{P,R} [\text{PAPR}(x, P, R)] \}$$
 [0145] where x are sequences for all possible extra tone values and rotations on the base sequence.

Note that PAPR only depends on product of P and R values [0146] $\{P,R\}=\{e.\text{sup}.j\phi, 1\}$ and $\{-e.\text{sup}.j\phi, -1\}$ gives the same sequence rotated by 180 degrees [0147] All possible values for $P*R$ for up to 8ss: $[1\ -1\ \exp(-j*\pi/3)\ \exp(-j*2*\pi/3)\ \exp(-j*4*\pi/3)\ \exp(-j*5*\pi/3)]$.

[0148] In one implementation, a punctured version of this EHT-LTF sequence (obtained from FIGS. 8, 10, 12, 14, and 16) may also be used may be used for 240 MHz transmission. In one aspect, the EHT-LTF sequence may be optimized for 240 MHz bandwidth channel using additional MRUs: 2×996 , $2\times 996+484$.

[0149] FIG. 18 is a block diagram illustrating an example of a wireless communication device that supports training fields for communications over large bandwidth channels that may achieve reduced peak-to-average power ratios. The wireless communication device 1800 may be, for example, an access point or a user station, and may be implemented with a processing system 1814 that includes one or more processors 1804. Examples of processors 1804 include microprocessors,

microcontrollers, digital signal processors (DSPs), field programmable gate arrays (FPGAs), programmable logic devices (PLDs), state machines, gated logic, discrete hardware circuits, and other suitable hardware configured to perform the various functionality described throughout this disclosure. In various examples, the wireless communication device **1800** may be configured to perform any one or more of the functions described herein. That is, the processor **1804**, as utilized in the wireless communication device **1800**, may be used to implement any one or more of the processes and procedures further illustrated in the flow diagrams of FIGS. **19** and **20**, to be discussed later.

[0150] In this example, the processing system **1814** may be implemented with a bus architecture, represented generally by the bus **1802**. The bus **1802** may include any number of interconnecting buses and bridges depending on the specific application of the processing system **1814** and the overall design constraints. The bus **1802** communicatively couples together various circuits including one or more processors (represented generally by the processor **1804**), a memory **1805**, and computer-readable media (represented generally by the computer-readable medium **1806**). The bus **1802** may also link various other circuits such as timing sources, peripherals, voltage regulators, and power management circuits, which are well known in the art, and therefore, will not be described any further. A bus interface **1808** provides an interface between the bus **1802** and a wireless transceiver **1810** (comprising a transmitter and a receiver). The wireless transceiver **1810** provides a communication interface or means for communicating with various other apparatus over a transmission medium. For instance, the wireless transceiver **1810** may transmit and receive to and from one or more wireless communication device using one or more antennas **1816** and in accordance with an IEEE **802.11** protocol, such as IEEE **802.11be**. In one implantation, the wireless transceiver **1814** may operate according to a multiple input multiple output (MIMO) mode.

[0151] The processor **1804** is responsible for managing the bus **1802** and general processing, including the execution of software stored on the computer-readable medium **1806**. The software, when executed by the processor **1804**, causes the processing system **1814** to perform the various functions described below for any particular apparatus. The computer-readable medium **1806** and the memory **1805** may also be used for storing data that is manipulated by the processor **1804** when executing software.

[0152] In one or more examples, the processor **1804** may include an OFDMA modulation circuit **1840**, an OFDMA demodulation circuit **1842**, and a Long Training Field Generator circuit **1844**. In one example, the Long Training Field Generator circuit **1844** may serve to obtain an EHT-LTF sequence for a 320 MHz bandwidth channel. The OFDMA modulation circuit **1840** may serve to modulate the EHT-LTF sequence and a data field into an OFDM signal for transmission. The OFDMA demodulation circuit **1842** may serve to demodulate a received OFDM signal.

[0153] One or more processors **1804** in the processing system may execute software. Software shall be construed broadly to mean instructions, instruction sets, code, code segments, program code, programs, subprograms, software modules, applications, software applications, software packages, routines, subroutines, objects, executables, threads of execution, procedures, functions, among other examples, whether referred to as software, firmware, middleware, microcode, hardware description language, or otherwise. The software may reside on a computer-readable medium **1806**. The computer-readable medium **1806** may be a non-transitory computer-readable medium. A non-transitory computer-readable medium includes, by way of example, a magnetic storage device (for example, hard disk, floppy disk, magnetic strip), an optical disk (for example, a compact disc (CD) or a digital versatile disc (DVD)), a smart card, a flash memory device (for example, a card, a stick, or a key drive), a random access memory (RAM), a read only memory (ROM), a programmable ROM (PROM), an erasable PROM (EPROM), an electrically erasable PROM (EEPROM), a register, a removable disk, and any other suitable medium for storing software or instructions that may be accessed and read by a computer. The computer-readable

medium **1806** may reside in the processing system **1814**, external to the processing system **1814**, or distributed across multiple entities including the processing system **1814**. The computer-readable medium **1806** may be embodied in a computer program product. By way of example, a computer program product may include a computer-readable medium in packaging materials. Those skilled in the art will recognize how best to implement the described functionality presented throughout this disclosure depending on the particular application and the overall design constraints imposed on the overall system.

[0154] In one or more examples, the computer-readable storage medium **1806** may include OFDMA modulation instructions **1850**, OFDMA demodulation instructions **1852**, or Long Training Fields instructions **1854**. Of course, in the above examples, the circuitry included in the processor **1804** is merely provided as an example, and other means for carrying out the described functions may be included within various aspects of the present disclosure, including but not limited to the instructions stored in the computer-readable storage medium **1806**, or any other suitable apparatus or means described in any one of the processes or algorithms described herein.

[0155] FIG. **19** is a flow chart illustrating an example method **1900** operational at a wireless communication device that supports training fields for communications over large bandwidth channels that may achieve reduced peak-to-average power ratios. At block **1902**, a data field is generated for transmission using a first channel having a 320 MHz total channel bandwidth or a 240 MHz total channel bandwidth.

[0156] At block **1904**, a long training field (LTF) sequence may be obtained (or retrieved, or generated) based on a concatenation of a plurality of LTF subsequences, each LTF subsequence being associated with a channel bandwidth less than the total channel bandwidth of the first channel, one or more of the plurality of LTF subsequences being selectively phase-rotated to reduce a peak-to-average power ratio.

[0157] In some implementations, the entire LTF sequence may be stored (in a storage device or memory device) at the wireless communication device in advance, such as during manufacturing or configuration. Then, the transmitting wireless communication device simply retrieves the entire LTF sequence from the memory device or storage device prior to transmission. That is, the LTF sequence need not be dynamically generated every time it is to be used.

[0158] In other implementations, the LTF sequence may be partially or fully generated or constructed at the transmitting wireless communication device, either dynamically or periodically. For example, the plurality of LTF subsequences may be stored (in a storage device or memory device) at the wireless communication device and the wireless communication device may generate the LTF sequence dynamically by applying a set of rotation coefficients or values (which are either stored at the wireless communication device or received) to the plurality of LTF subsequences, and then concatenating the phase-rotated LTF subsequences. In still other implementations, pre-phase-rotated LTF subsequences may be stored in the wireless communication device, which may then be dynamically or periodically concatenated at the wireless communication device prior to transmission of the LTF sequence. In yet other implementations, a plurality of base sequences are stored at the wireless communication device. In such implementations, the wireless communication device may dynamically retrieve the base sequences from memory, construct the LTF subsequences, phase-rotate the LTF subsequences, and concatenate the LTF subsequences to construct the LTF sequence.

[0159] In one example, each of the plurality of LTF subsequences is based on High Efficiency (HE) LTF (HE-LTF) sequence for a subchannel having an 80 MHz channel bandwidth. For example, the LTF sequence may be one of the EHT-LTF sequences generated according to FIGS. **8**, **10**, **12**, and **14**.

[0160] At block **1906**, a wireless packet may be transmitted that includes the LTF and the data field to at least one second wireless communication device via the first channel.

[0161] In one implementation, a symbol duration of each symbol corresponding to the LTF

sequence is one-fourth as long as a symbol duration of each symbol of the data field, and where the concatenation is a sequence formed by concatenating, in any order, a first LTF subsequence of the plurality of LTF subsequences, a second LTF subsequence of the plurality of LTF subsequences, a third LTF subsequence of the plurality of LTF subsequences, and a fourth LTF subsequence of the plurality of LTF subsequences, with k filler zeros between adjacent ones of the plurality of LTF subsequences. For instance, the symbol duration of each symbol of the LTF sequence may be $3.2\ \mu\text{s}$ plus a guard interval (GI) and the symbol duration of each symbol of the data field may be $12.8\ \mu\text{s}$ plus the guard interval. In some implementation, the GI may be $0.8\ \mu\text{s}$, or $1.6\ \mu\text{s}$. In this example, the LTF sequence may be generated in accordance with FIGS. **8** and **9** where the phase rotation values s are selected to reduce or minimize a PAPR for the LTF transmission.

[0162] In another implementation, a symbol duration of each symbol corresponding to the LTF sequence is half as long as a symbol duration of each symbol of the data field, and the concatenation is a sequence formed by concatenating, in any order, a first LTF subsequence of the plurality of subsequences, a second LTF subsequence of the plurality of LTF subsequences, a third LTF subsequence of the plurality of LTF subsequences, and a fourth LTF subsequence of the plurality of LTF subsequences, with k filler zeros between adjacent ones of the plurality of LTF subsequences. For instance, the symbol duration of each symbol of the LTF sequence may be $6.4\ \mu\text{s}$ plus a guard interval (GI) and the symbol duration of each symbol of the data field may be $12.8\ \mu\text{s}$ plus the guard interval. In some implementation, the GI may be $0.8\ \mu\text{s}$, or $1.6\ \mu\text{s}$. In this example, the LTF sequence may be generated in accordance with FIGS. **10** and **11** where the phase rotation values q are selected to reduce or minimize a PAPR for the LTF transmission.

[0163] In yet another implementation, a symbol duration of each symbol corresponding to the LTF sequence is the same as a symbol duration of each symbol of the data field, and where the concatenation is a sequence formed by concatenating, in any order, a first LTF subsequence of the plurality of LTF subsequences, a second LTF subsequence of the plurality of LTF subsequences, a third LTF subsequence of the plurality of LTF subsequences, and a fourth LTF subsequence of the plurality of LTF subsequences, with k filler zeros between adjacent ones of the plurality of LTF subsequences. For instance, the symbol duration of each symbol of the LTF sequence may be $12.8\ \mu\text{s}$ plus a guard interval (GI) and the symbol duration of each symbol of the data field may be $12.8\ \mu\text{s}$ plus the guard interval. In some implementation, the GI may be $0.8\ \mu\text{s}$, or $3.2\ \mu\text{s}$. In this example, the LTF sequence may be generated in accordance with FIGS. **12** and **13** or FIGS. **14** and **15** where the phase rotation values n are selected to reduce or minimize a PAPR for the LTF transmission.

[0164] FIG. **20** is a flow chart illustrating an example method **2000** operational at a wireless communication device that supports training fields for communications over large bandwidth channels that may achieve reduced peak-to-average power ratios. The method illustrated in FIG. **20** may receive the LTF sequence(s) illustrated in FIGS. **8-17** as part of a transmission.

[0165] At block **2002**, a wireless packet is received that includes a long training field (LTF) sequence and a data field via a first channel having a 320 MHz total channel bandwidth or a 240 MHz total channel bandwidth.

[0166] At block **2004**, the LTF sequence is detected within the packet, the LTF sequence based on a concatenation of a plurality of LTF subsequences, each LTF subsequence being associated with a channel bandwidth less than the total channel bandwidth of the first channel, one or more of the plurality of LTF subsequences being selectively phase-rotated to reduce a peak-to-average power ratio.

[0167] At block **2006**, the data field is received based on the detected LTF sequence. The following provides an overview of aspects of the present disclosure:

[0168] Aspect 1: A method for wireless communication by a wireless communication device, comprising generating a data field for transmission using a first channel having a 320 MHz total channel bandwidth or a 240 MHz total channel bandwidth; obtaining a long training field (LTF)

0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, +1, 0, 0, 0, +1}.

[0176] Aspect 9: The method of aspect 1 or 2, wherein a symbol duration of each symbol of the LTF sequence is the same as a symbol duration of each symbol of the data field, and the concatenation is a sequence formed by concatenating, in any order, a first LTF subsequence of the plurality of LTF subsequences, a second LTF subsequence of the plurality of LTF subsequences, a third LTF subsequence of the plurality of LTF subsequences, and a fourth LTF subsequence of the plurality of LTF subsequences, with k filler zeros between adjacent ones of the plurality of LTF subsequences.

[0177] Aspect 10: The method of aspect 1, 2, or 9, wherein the selectively phase-rotated one or more of the plurality of LTF subsequences are based on a set of phase rotation values $n=[n_1\ n_2\ n_3\ n_4\ n_5\ n_6\ n_7\ n_8]$ to the plurality of LTF subsequences, each of the LTF subsequences being based on a first LTF base sequence LTF-A and a second LTF base sequence LTF-B, and the first LTF subsequence= $\{n_1*\text{LTF-A}, 0, n_2*\text{LTF-B}\}$; the second LTF subsequence= $\{n_3*\text{LTF-A}, 0, n_4*\text{LTF-B}\}$; the third LTF subsequence= $\{n_5*\text{LTF-A}, 0, n_6*\text{LTF-B}\}$; and the fourth LTF subsequence= $\{n_7*\text{LTF-A}, 0, n_8*\text{LTF-B}\}$.

[0178] Aspect 11: A wireless communication device, comprising: at least one processor; and at least one memory communicatively coupled with the at least one processor and storing processor-readable code that, when executed by the at least one processor, is configured to: generate a data field for transmission using a first channel having a 320 MHz total channel bandwidth or a 240 MHz total channel bandwidth; obtaining a long training field (LTF) sequence based on a concatenation of a plurality of LTF subsequences, each LTF subsequence being associated with a channel bandwidth less than the total channel bandwidth of the first channel, one or more of the plurality of LTF subsequences being selectively phase-rotated to reduce a peak-to-average power ratio; and transmit a wireless packet that includes the LTF sequence and the data field to at least one second wireless communication device via the first channel.

[0179] Aspect 12: The wireless communication device of aspect 11, wherein a symbol duration of each symbol of the LTF sequence is one-fourth as long as a symbol duration of each symbol of the data field, and wherein the concatenation is a sequence formed by concatenating, in any order, a first LTF subsequence of the plurality of LTF subsequences, a second LTF subsequence of the plurality of LTF subsequences, a third LTF subsequence of the plurality of LTF subsequences, and a fourth LTF subsequence of the plurality of LTF subsequences, with k filler zeros between adjacent ones of the plurality of LTF subsequences.

[0180] Aspect 13: The wireless communication device of any one of aspects 11 or 12, wherein the selectively phase-rotated one or more of the plurality of LTF subsequences are based on a set of phase rotation values $s=[s_1\ s_2\ s_3\ s_4\ s_5\ s_6\ s_7\ s_8]$ to the plurality of LTF subsequences, each of the LTF subsequences being based on a first LTF base sequence LTF-A and a second LTF base sequence LTF-B, and the first LTF subsequence= $\{s_1*\text{LTF-A}, 0, s_2*\text{LTF-B}\}$; the second LTF subsequence= $\{s_3*\text{LTF-A}, 0, s_4*\text{LTF-B}\}$; the third LTF subsequence= $\{s_5*\text{LTF-A}, 0, s_6*\text{LTF-B}\}$; and the fourth LTF subsequence= $\{s_7*\text{LTF-A}, 0, s_8*\text{LTF-B}\}$.

[0181] Aspect 14: The wireless communication device of aspect 13, wherein the set of phase rotation values $s=[s_1\ s_2\ s_3\ s_4\ s_5\ s_6\ s_7\ s_8]=[1\ 1\ 1\ 1\ -1\ -1\ -1\ -1]$.

[0182] Aspect 15: The wireless communication device of any one of aspects 12, 13, or 14, wherein

understood that the described program components and systems can generally be integrated together in a single software product or packaged into multiple software products.

Claims

1-20. (canceled)

21. A wireless communication device, comprising: at least one processor; and at least one memory communicatively coupled with the at least one processor and storing processor-readable code that, when executed by the at least one processor, is configured to: generate a data field for transmission using a first channel having a 320 MHz total channel bandwidth or a 240 MHz total channel bandwidth; obtain a long training field (LTF) sequence based on a concatenation of a plurality of LTF subsequences, each LTF subsequence being associated with a channel bandwidth less than the total channel bandwidth of the first channel, wherein a symbol duration of each symbol of the LTF sequence is one-fourth as long as a symbol duration of each symbol of the data field, and wherein the concatenation is a sequence formed by concatenating, in any order, a first LTF subsequence of the plurality of LTF subsequences, a second LTF subsequence of the plurality of LTF subsequences, a third LTF subsequence of the plurality of LTF subsequences, and a fourth LTF subsequence of the plurality of LTF subsequences, with k filler zeros between adjacent ones of the plurality of LTF subsequences; and transmit a wireless packet that includes the LTF sequence and the data field to at least one second wireless communication device via the first channel.

22. The wireless communication device of claim 21, wherein each of the plurality of LTF subsequences is based on a High Efficiency (HE) LTF (HE-LTF) sequence for a subchannel having an 80 MHz channel bandwidth.

23. The wireless communication device of claim 21, wherein the symbol duration of each symbol of the LTF sequence is 3.2 μ s plus a guard interval and the symbol duration of each symbol of the data field is 12.8 μ s plus the guard interval.

24. The wireless communication device of claim 21, wherein one or more of the plurality of LTF subsequences are phase-rotated.

25. The wireless communication device of claim 4, wherein the phase-rotated one or more of the plurality of LTF subsequences are based on a set of phase rotation values $s=[s_1 \ s_2 \ s_3 \ s_4 \ s_5 \ s_6 \ s_7 \ s_8]$ to the plurality of LTF subsequences, where each rotation value $s_1, s_2, s_3, s_4, s_5, s_6, s_7, s_8$ is either +1, 0, or -1, each of the plurality of LTF subsequences being based on a first LTF base sequence LTF-A and a second LTF base sequence LTF-B, the first and second LTF base sequences LTF-A and LTF-B including a plurality of +1, 0, and/or -1 values.

26. The wireless communication device of claim 25, wherein: the first LTF subsequence= $\{s1*\text{LTF-A}, 0, s2*\text{LTF-B}\}$; the second LTF subsequence= $\{s3*\text{LTF-A}, 0, s4*\text{LTF-B}\}$; the third LTF subsequence= $\{s5*\text{LTF-A}, 0, s6*\text{LTF-B}\}$; and the fourth LTF subsequence= $\{s7*\text{LTF-A}, 0, s8*\text{LTF-B}\}$.

27. The wireless communication device of claim 26, wherein the set of phase rotation values $s=[s_1 \ s_2 \ s_3 \ s_4 \ s_5 \ s_6 \ s_7 \ s_8]=[1 \ 1 \ 1 \ 1 \ -1 \ -1 \ -1 \ -1]$.

[illegible]

subsequences is based on a High Efficiency (HE) LTF (HE-LTF) sequence for a subchannel having an 80 MHz channel bandwidth.

37. The wireless communication device of claim 35, wherein the symbol duration of each symbol of the LTF sequence is $3.2\ \mu\text{s}$ plus a guard interval and the symbol duration of each symbol of the data field is $12.8\ \mu\text{s}$ plus the guard interval.

38. The wireless communication device of claim 35, wherein one or more of the plurality of LTF subsequences are phase-rotated.

39. The wireless communication device of claim 38, wherein the phase-rotated one or more of the plurality of LTF subsequences are based on a set of phase rotation values $s=[s_1\ s_2\ s_3\ s_4\ s_5\ s_6\ s_7\ s_8]$, where each rotation value $s_1, s_2, s_3, s_4, s_5, s_6, s_7, s_8$ is either +1, 0, or -1, and each of the plurality of LTF subsequences is based on a first LTF base sequence LTF-A and a second LTF base sequence LTF-B, the first and second LTF base sequences LTF-A and LTF-B including a plurality of +1, 0, and/or -1 values.

40. The wireless communication device of claim 39, wherein: the first LTF subsequence= $\{s_1*\text{LTF-A}, 0, s_2*\text{LTF-B}\}$; the second LTF subsequence= $\{s_3*\text{LTF-A}, 0, s_4*\text{LTF-B}\}$; the third LTF subsequence= $\{s_5*\text{LTF-A}, 0, s_6*\text{LTF-B}\}$; and the fourth LTF subsequence= $\{s_7*\text{LTF-A}, 0, s_8*\text{LTF-B}\}$.
